

Negative Stepsizes Make Gradient-Descent-Ascent Converge

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Abstract

Efficient computation of min-max problems is a central question in optimization, learning, games, and controls. Arguably the most natural algorithm is gradient-descent-ascent (GDA). However, since the 1970s, conventional wisdom has argued that GDA fails to converge even on simple problems. This failure spurred an extensive literature on modifying GDA with additional building blocks such as extragredients, optimism, momentum, anchoring, etc. In contrast, we show that GDA converges in its original form by simply using a judicious choice of stepsizes.

The key innovation is the proposal of unconventional stepsize schedules (dubbed *slingshot stepsize schedules*) that are time-varying, asymmetric, and periodically negative. We show that all three properties are necessary for convergence, and that altogether this enables GDA to converge on the classical counterexamples (e.g., unconstrained convex-concave problems). All of our results apply to the last iterate of GDA, as is typically desired in practice.

The core algorithmic intuition is that although negative stepsizes make backward progress, they de-synchronize the min and max variables (overcoming the cycling issue of GDA), and lead to a *slingshot phenomenon* in which the forward progress in the other iterations is overwhelmingly larger. This results in fast overall convergence. Crucially, for this de-synchronization we break symmetry by alternating negative steps for the min and max variables—this goes beyond what is possible with classical reductions from min-max problems to monotone operator theory, highlighting the importance of the intrinsic asymmetry in min-max problems.

Geometrically, the slingshot dynamics leverage the non-reversibility of gradient flow: positive/negative steps cancel to first order, yielding a second-order net movement in a new direction that leads to convergence and is otherwise impossible for GDA to move in. We interpret this as a second-order finite-differencing algorithm and show that, intriguingly, it approximately implements consensus optimization, an empirically popular algorithm for min-max problems involving deep neural networks (e.g., training GANs).

Contents

1	Introduction	3
1.1	Contribution: GDA actually can converge	4
1.2	Slingshot stepsize schedules	4
1.3	Plausibility argument for bilinear problems	7
1.4	Interpretation via second-order approximation	9
1.5	Related work	10
2	Preliminaries	14
2.1	Min-max optimization	14
2.2	Failure of GDA with standard stepsizes	15
2.3	Chebyshev polynomials	16
3	Min-max problems with linear gradients	16
3.1	Bilinear objectives	17
3.2	Quadratic objectives	19
3.3	Alternative implementations and complex stepsizes	22
4	Min-max problems with nonlinear gradients	22
4.1	Convex-concave objectives	23
4.2	Strongly-convex-strongly-concave objectives	24
4.3	Proof of 2-step lemma	25
5	Connections to consensus optimization and finite differencing	27
5.1	Linear version of the slingshot stepsize schedule	27
5.2	Nonlinear version of the slingshot stepsize schedule	29
6	Discussion	30
A	Alternative implementations	33
A.1	Random stepsizes	33
A.2	Stable fractal stepsizes	34
B	Deferred details	35
B.1	General second-order approximations	35
B.2	Proof of Lemma 2.2	35
B.3	Proof of Theorem 3.4	37
B.4	Proof of Lemma 3.6	38
B.5	Experimental setup	38
	References	38

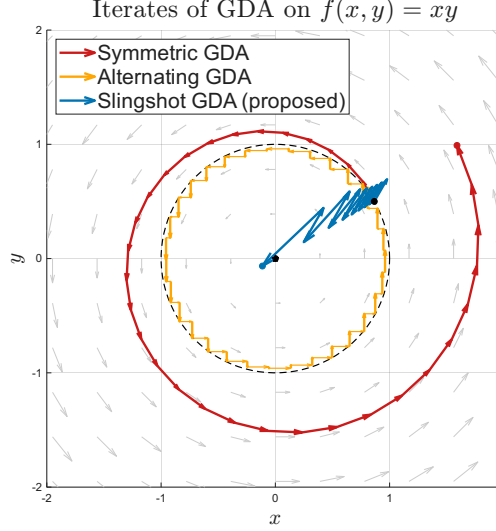


FIGURE 1: GDA trajectories for a classical counterexample: the unconstrained bilinear problem $\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} xy$. The unique solution is the origin. GDA fails to converge with standard stepsize schedules, e.g., it diverges if $\alpha_t = \beta_t > 0$ (red), or limit-cycles if α_t, β_t are alternately positive and zero (yellow). Our proposed stepsize schedule (blue) enables GDA to converge for the first time. Standard GDA attempts to follow the grey vector field $(-\nabla_x f, \nabla_y f) = (-y, x)$; in contrast, our proposed GDA alternately moves in the directions $(\nabla_x f, \nabla_y f) = (y, x)$ and $(-\nabla_x f, -\nabla_y f) = (-y, -x)$. This trajectory “slingshots” to make progress every two iterations.

1 Introduction

This paper revisits the numerical computation of min-max problems, a.k.a. saddle-point problems:

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y). \quad (1.1)$$

Efficient computation of such problems is a central question in many areas, such as applied mathematics [12], game theory [85], constrained optimization [13], distributed optimization [17], robust optimization [11], robust control [34], adversarial machine learning [57], and generative adversarial networks (GANs) [30], among many others.

First-order algorithms are the predominant choice for solving (1.1) due to their scalability, simplicity, and flexibility. Arguably the most natural such algorithm is gradient-descent-ascent (GDA), which is the extension of gradient descent from standard optimization to min-max optimization:

$$\begin{aligned} x_{t+1} &= x_t - \alpha_t \nabla_x f(x_t, y_t), \\ y_{t+1} &= y_t + \beta_t \nabla_y f(x_t, y_t). \end{aligned} \quad (1.2)$$

However, conventional wisdom states that GDA fails to converge even on simple problems. A classical counterexample dating at least to Korpelevich in 1976 [41] is that GDA either diverges or gets stuck in limit cycles even on the toy 1-dimensional unconstrained bilinear problem

$$\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} xy. \quad (1.3)$$

Indeed, for this problem, the GDA update (1.2) simplifies to a linear dynamical system

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -\alpha_t \\ \beta_t & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} \quad (1.4)$$

that is expansive since the determinant of the update matrix is $1 + \alpha_t \beta_t$, which is at least 1 for any standard choice of GDA stepsizes $\alpha_t, \beta_t \geq 0$. See Figure 1 for an illustration, and see §2.2 for full details on this counterexample.

In fact, this failure is generic: the last iterate of GDA fails to converge on *every* bilinear problem, not just this toy problem (1.3). Moreover, this failure provably applies to arbitrary stepsize schedules $\alpha_t, \beta_t \in \mathbb{R}$ which fall under any of the standard categories: non-negative stepsizes ($\alpha_t, \beta_t \geq 0$), symmetric stepsizes ($\alpha_t = \beta_t$), and/or constant stepsizes ($\alpha_t \equiv \alpha, \beta_t \equiv \beta$).

There are only a few restricted settings where GDA converges: convex-concave problems with uniformly bounded subgradients [62, 95], and under strong growth conditions such as strong convexity and strong concavity (albeit at a provably slow convergence rate [48, 51]). These assumptions prohibit natural settings like unconstrained bilinear, quadratic, or convex-concave problems. Moreover, the former result only applies to the *average* iterate—since even in the toy bilinear problem $\min_{x:|x|\leq 1} \max_{y:|y|\leq 1} xy$, the iterates still cycle/diverge, constrained only by the compact domain (imagine Figure 1 with a bounding box). This is a well-documented issue for applications like training GANs where convexity-concavity fails and average iterates are nonsensical models.

These important issues have spurred a fruitful and extensive literature aimed at obtaining last-iterate convergence without strong assumptions. Since the seminal 1976 paper of Korpelevich [41], the de facto approach has been to modify GDA with additional building blocks such as extragredients, momentum, optimism, anchoring, and more; see the prior work section below.

1.1 Contribution: GDA actually can converge

Contrary to conventional wisdom, we show that GDA actually can converge in its original form (1.2) by judiciously choosing stepsizes. The key innovation is the proposal of unconventional stepsize schedules—dubbed *slingshot stepsize schedules*, described below—that are *time-varying*, *asymmetric*, and (most surprisingly) use interspersed *negative stepsizes*. All three characteristics are provably necessary for GDA to converge even on the bilinear counterexample (1.3).

Our main results prove that these stepsize schedules enable the last iterate of GDA to converge on unconstrained bilinear, quadratic, and convex-concave min-max problems—the classical counterexamples for GDA. Our results do not require extragredients, momentum, optimism, anchoring, or any other modification to GDA.

We emphasize that the focus of this paper is on whether GDA can converge at all and why our counterintuitive stepsizes are helpful, not on obtaining the fastest complexity bounds for any algorithm—if that were the goal, it would be sensible to incorporate additional building blocks. Nevertheless, we mention that the resulting convergence rates are competitive with specially tailored algorithms. For bilinear and quadratic problems, our stepsize schedule enables GDA to converge at an accelerated rate which exactly matches the information-theoretically optimal rate among arbitrary first-order methods. For convex-concave problems, our stepsize schedule enables GDA to match the known convergence rates for extragradient and optimistic GDA. Finally, for strongly-convex-strongly-concave problems, GDA already converges with standard stepsize schedules (this is the only such setting in the paper) but at a slow rate; our stepsize schedule enables GDA to converge at the optimal accelerated rate for the first time.

1.2 Slingshot stepsize schedules

Since optimal implementations vary across settings, we propose a *family* of stepsize schedules. These schedules pair stepsizes in consecutive iterations $2t, 2t + 1$ so that:

- (i) Negative stepsizes: At least one of $\alpha_{2t}, \alpha_{2t+1}, \beta_{2t}, \beta_{2t+1}$ is strictly negative.

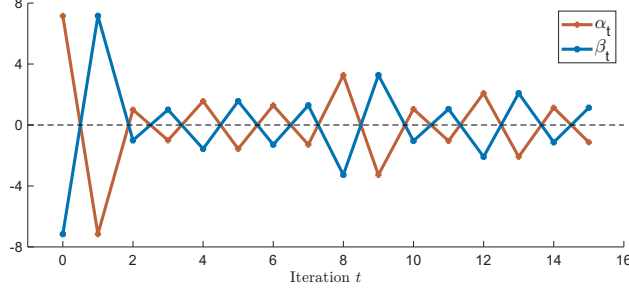


FIGURE 2: Optimal stepsize schedule for unconstrained bilinear min-max optimization, here for 16 iterations, for a bilinear coupling with spectral bounds $1/100 \leq \sigma_i^2(\nabla_{xy}^2 f) \leq 1$, and in the stable fractal ordering discussed in [Appendix A.2](#). Recall that α_t and β_t respectively denote the stepsizes for x and y ; standard schedules take $\alpha_t, \beta_t \geq 0$ and often equal. Observe that the proposed stepsizes are time-varying, asymmetric, and alternately negative.

(ii) Alternating products: $\alpha_{2t}\beta_{2t+1} \geq 0$ and $\alpha_{2t+1}\beta_{2t} \geq 0$.

(iii) Consecutive sums: $\alpha_{2t} + \alpha_{2t+1} \geq 0$ and $\beta_{2t} + \beta_{2t+1} \geq 0$.

Property (iii) ensures that the forward movement outweighs the backward movement from (i), and (ii) ensures that the higher-order net movement arising from positive/negative cancellations is in a helpful direction. We describe this in detail below; for concreteness we first start with two examples.

Concrete examples. In this paper we focus on two instantiations of this general family of stepsize schedules, dichotomized by whether the min-max objective f has a linear gradient ∇f :

- Linear ∇f . For such problems (i.e., bilinear and quadratic objectives), we propose slingshot stepsize schedules of the form

$$\alpha_{2t} = -\beta_{2t} = -\alpha_{2t+1} = \beta_{2t+1} = h_t \quad (1.5)$$

for a suitable choice of $\{h_t\}$. Observe that the minimization variable x performs a positive step ($\alpha_{2t} > 0$) followed by a negative step ($\alpha_{2t+1} < 0$) if $h_t > 0$, and vice versa for the maximization variable y . Thus, in each iteration, the stepsizes for x and y are *anti-correlated* in that they are equal and opposite. See [Figure 2](#) for an illustration.

- Nonlinear ∇f . For such problems (e.g., convex-concave objectives), we propose slingshot stepsize schedules of the form

$$\alpha_{2t} = -\beta_{2t} = \beta_{2t+1} = h, \alpha_{2t+1} = 0 \quad \text{or} \quad -\alpha_{2t} = \beta_{2t} = \alpha_{2t+1} = h, \beta_{2t+1} = 0, \quad (1.6)$$

where we randomize this “or” with probability 1/2 to break the symmetry between x and y . Notice that properties (i)-(iii) hold almost surely, i.e., in every realization of the stepsizes.

The reason we study multiple versions of the slingshot stepsize schedule is that the former proposal is exactly optimal along all stepsize schedules for settings where ∇f is linear (see [§3](#)), but actually fails to converge in settings where ∇f is nonlinear (see [§4](#)). The latter proposal fixes this convergence issue for general convex-concave problems.

Three key properties. Slingshot stepsize schedules are time-varying, asymmetric, and periodically use negative stepsizes. As already mentioned, all three properties are provably necessary¹ for GDA to converge even on bilinear problems. We comment on each of these three key properties:

- Negative stepsizes (i.e., $\alpha_t < 0$ or $\beta_t < 0$). Such steps move *backward*: they move in the positive gradient direction (an ascent direction) for the minimization variable x , and in the negative gradient direction (a descent direction) for the maximization variable y . At first glance, it seems that such steps are nonsensical and should only worsen the standard divergent behavior of GDA. Counterintuitively, we show that such steps are essential for convergence. Negative steps make backward progress, but de-synchronize the min and max variables (overcoming the cycling issue of GDA), and lead to a *slingshot phenomenon* in which the forward progress in the other iterations (with positive stepsizes) is overwhelmingly larger. This results in fast overall convergence. The key intuition is that the positive and negative movements cancel to first order, resulting in a second-order movement in a new direction that leads to convergence and is otherwise impossible for GDA to move in. We explain this in detail in §1.4.
- Asymmetric stepsizes (i.e., $\alpha_t \neq \beta_t$). Slingshot stepsize schedules update differently the minimization variable x and maximization variable y . This goes beyond what is possible with classical reductions from min-max problems to monotone operator theory (which always concatenate the variables $z = (x, y)$ and then treat them symmetrically), highlighting the importance of the intrinsic asymmetry in min-max problems.

In fact, fundamental obstructions prohibit extending the slingshot stepsizes from convex-concave min-max problems to variational inequalities with monotone operators. Taking different stepsizes in x and y is not even *implementable* for generic monotone operator problems since operator monotonicity is invariant with respect to coordinate permutations and in particular does not have block structure $z = (x, y)$ as in min-max problems. Further, taking the same stepsizes in all variables—i.e., algorithms of the form $z_{t+1} = z_t - \eta_t F(z_t)$ —fails to solve unconstrained variational inequalities $F(z) = 0$ for smooth monotone operators F for any choice of stepsizes η_t ; indeed, in the special case of min-max problems, such algorithms devolve into GDA with symmetric stepsizes, which fails to converge even for smooth bilinear problems (see §2.2). Our results exploit the block structure of the variables $z = (x, y)$ to implement the slingshot stepsizes, and exploit the convex-concave block structure of the monotone operator $F(z) = (\nabla_x f, -\nabla_y f)$ to analyze the slingshot stepsizes.

- Time-varying stepsizes (i.e., α_t, β_t are not constant in t). Slingshot stepsize schedules do not make progress in each iteration (negative steps make backward progress). Nevertheless, we prove that this algorithm converges in the long-run.

We remark that slingshot stepsize schedules can be time-varying in two ways. Varying *within* pairs of iterations (e.g., $\alpha_{2t} \neq \alpha_{2t+1}$ and $\beta_{2t} \neq \beta_{2t+1}$) enables GDA to converge. Further varying *between* pairs of iterations (i.e., h_t varying in t) can accelerate this convergence.

While some existing stepsize schedules have the latter two properties (diminishing stepsizes are time-varying, two-timescale GDA is asymmetric, and alternating GDA is time-varying and asymmetric), none have the first property. Recall that *all* three properties are provably required for

¹This statement applies to *real* stepsize schedules $\alpha_t, \beta_t \in \mathbb{R}$. Intriguingly, we show that *complex* stepsize schedules can also make GDA converge (and this does not require asymmetry); details in §3.3. This appears to be the first use of complex stepsizes in optimization or min-max optimization, and is surprising because the problem data is purely real. However, these complex stepsizes are specially tailored to the unconstrained bilinear setting and do not extend gracefully to more general settings, hence we focus on real stepsizes. A thorough investigation of complex stepsizes will appear in forthcoming work.

GDA to converge. Perhaps the closest similarity to negative stepsizes is negative momentum [27], but that is fundamentally different because it changes GDA by adding internal dynamics, uses positive stepsizes, and often has time-invariant, symmetric parameters. Moreover, it is unknown if negative momentum can accelerate in bilinear or strongly-convex-strongly-concave settings or can even converge in convex-concave settings, whereas we show that all of this is possible using the slingshot stepsizes. See the related work §1.5 for further discussion.

1.3 Plausibility argument for bilinear problems

Here we provide a simple plausibility argument for why time-varying, asymmetric, and periodically negative stepsizes enable GDA to converge. For this argument, consider bilinear problems $\min_x \max_y x^\top B y$ with stationary solution at the origin (without loss of generality by translation), and for simplicity assume that B has non-zero singular values (so that $m\mathbf{I} \leq BB^\top, B^\top B \leq M\mathbf{I}$); the general case is a straightforward extension via SVD, see §3.1.

A first convergence result. For bilinear problems, two iterations of the GDA algorithm (1.2) amounts to the linear dynamical system

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} \mathbf{I} & -\alpha_1 B \\ \beta_1 B^\top & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} & -\alpha_0 B \\ \beta_0 B^\top & \mathbf{I} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad (1.7)$$

Consider using the slingshot stepsize schedule (1.5), that is

$$\alpha_0 = \beta_1 = h \quad \text{and} \quad \beta_0 = \alpha_1 = -h, \quad (1.8)$$

for $h > 0$. Then this two-step update simplifies to

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I} & hB \\ hB^\top & \mathbf{I} \end{bmatrix}}_{\mathbf{U}_1} \underbrace{\begin{bmatrix} \mathbf{I} & -hB \\ -hB^\top & \mathbf{I} \end{bmatrix}}_{\mathbf{U}_0} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I} - h^2 BB^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{I} - h^2 B^\top B \end{bmatrix}}_{\mathbf{U}_1 \mathbf{U}_0} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}, \quad (1.9)$$

which is contractive $\|\mathbf{U}_1 \mathbf{U}_0\| < 1$ for sufficiently small $h > 0$ since $BB^\top$ and $B^\top B$ are positive definite. It immediately follows that repeating this 2-step schedule (1.8)—that is, $\alpha_n = h, \beta_n = -h$ for n even, and vice versa for n odd—makes GDA converge, and moreover at an exponential rate.

Notice that the key properties of the stepsizes (1.8) used for the two-step contraction (1.9) are that $\alpha_0 + \alpha_1 = \beta_0 + \beta_1 = 0$ (so that the off-diagonal blocks of $\mathbf{U}_1 \mathbf{U}_0$ vanish) and $\alpha_0 \beta_1 = \alpha_1 \beta_0 > 0$ (so that the diagonal blocks of $\mathbf{U}_1 \mathbf{U}_0$ are contractive). These two properties correspond respectively to (iii) and (ii) in the definition of slingshot stepsize schedules in §1.2.

Key point: contraction in two iterations, despite expansion in any single iteration.

To understand this result, it is important to emphasize that GDA converges (i.e., $\|\mathbf{U}_1 \mathbf{U}_0\| < 1$) even though each individual iteration is expansive in some directions (i.e., $\|\mathbf{U}_1\|, \|\mathbf{U}_0\| > 1$).² The expansiveness of each individual step is due to the backward movement: the negative stepsizes $\alpha_1 < 0$ and $\beta_0 < 0$ respectively amount to an ascent step in the minimization variable x and a descent step in the maximization variable y . Yet, overall convergence occurs because a variable’s backward movement in one iteration is outweighed by the forward movement in the paired iteration. This is the “slingshot” behavior illustrated in Figure 1.

²The expansiveness is because both $\mathbf{U}_0$ and $\mathbf{U}_1$ can have norm $1 + h\sqrt{M} > 1$ even for 1-dimensional problems; consider the eigenvectors $[1, 1]$ and $[1, -1]$ and the instances $B = -\sqrt{M}$ or $\sqrt{M}$, respectively.

This can be intuitively understood via the following 2×2 example:

$$\begin{bmatrix} 1 & \varepsilon \\ \varepsilon & 1 \end{bmatrix} \begin{bmatrix} 1 & -\varepsilon \\ -\varepsilon & 1 \end{bmatrix} = \begin{bmatrix} 1 - \varepsilon^2 & 0 \\ 0 & 1 - \varepsilon^2 \end{bmatrix} \quad (1.10)$$

has norm smaller than 1, even though both constituent matrices have norm larger than 1. This occurs because the expansive eigenvectors $(1, 1)$ and $(1, -1)$ for $\mathbf{U}_1$ and $\mathbf{U}_0$, respectively, are contractive eigenvectors for the opposite update matrix, and the contraction $1 - \varepsilon$ outweighs the expansion $1 + \varepsilon$ since $(1 - \varepsilon)(1 + \varepsilon) = 1 - \varepsilon^2 < 1$.

In fact, this simple example (1.10) is the crux of why $\|\mathbf{U}_1 \mathbf{U}_0\| < 1$ in the two-step update (1.9): in 1 dimension, $\mathbf{U}_0$ and $\mathbf{U}_1$ are of this form modulo permutation; and in general dimension, simultaneous diagonalization reduces $\mathbf{U}_0$ and $\mathbf{U}_1$ to this form on each eigenspace.

Optimizing the convergence rate. So far we have shown that for sufficiently small $h > 0$, periodically repeating the two-step schedule (1.8) makes GDA converge at an exponential rate. This amounts to the slingshot stepsize schedule (1.5) with $h_t = h$ for all t . Now that convergence is ensured, it is natural to ask how fast can one make this exponential convergence? How small should h be? What is the optimal stepsize schedule? Should the optimal h_t vary over time?

Conveniently, all of these questions are answered by appealing to the following simple but key interpretation of (1.9). We emphasize that this reduction applies only to the slingshot stepsize schedule and not to any other existing schedules.

Observation 1.1 (Paired GDA slingshot steps are equivalent to GD steps on the Hamiltonian). *Suppose f is bilinear. Then the following are equivalent:*

- Two iterations of GDA with the slingshot stepsize schedule (1.8) on $\min_x \max_y f(x, y)$.
- One iteration of GD with stepsize h^2 on $\min_{x, y} \frac{1}{2} \|\nabla f(x, y)\|^2$.

Indeed, a simple calculation shows that one iteration of GD on the “Hamiltonian” $\Phi(x, y) := \frac{1}{2} \|\nabla f(x, y)\|^2 = \frac{1}{2} x^\top \mathbf{B} \mathbf{B}^\top x + \frac{1}{2} y^\top \mathbf{B}^\top \mathbf{B} y$ is equivalent to the update (1.9). The upshot is that the Hamiltonian is a smooth, strongly-convex quadratic ($m\mathbf{I} \leq \nabla^2 \Phi \leq M\mathbf{I}$), hence we may appeal to classical stepsize schedules for GD. For example, simply repeating (1.9) with $h = \sqrt{2/(M+m)}$ yields exponential convergence with contraction factor $(\kappa - 1)/(\kappa + 1)$, where $\kappa := M/m$ denotes the condition number. Better yet, using optimal GD stepsize schedules [89] improves this to the accelerated rate $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$. This is our proposed slingshot stepsize schedule (1.5) for bilinear problems, which uses time-varying h_t based on the roots of Chebyshev polynomials. See §3.1 for full details on this stepsize schedule and the fact that it not only converges, but moreover converges at the accelerated rate that is information-theoretically optimal among first-order methods. See also Figure 3 for an illustration of how the practical performance matches the theory.

Beyond bilinear problems. The above calculations can be extended from bilinear to quadratic objectives, essentially since linearity of ∇f is the key property needed for the two-step argument (1.9); details in §3. However, further analysis techniques are required beyond the setting of linear ∇f , e.g., for convex-concave f . Of course one could try to linearize and then use a similar spectral analysis, but such arguments would be geared toward local rather than global convergence and moreover would necessitate assuming much stronger structural properties on f than are necessary. In §4 we prove progress after two steps for the standard setting where f is only assumed convex-concave and smooth, and then use that to conclude global convergence.

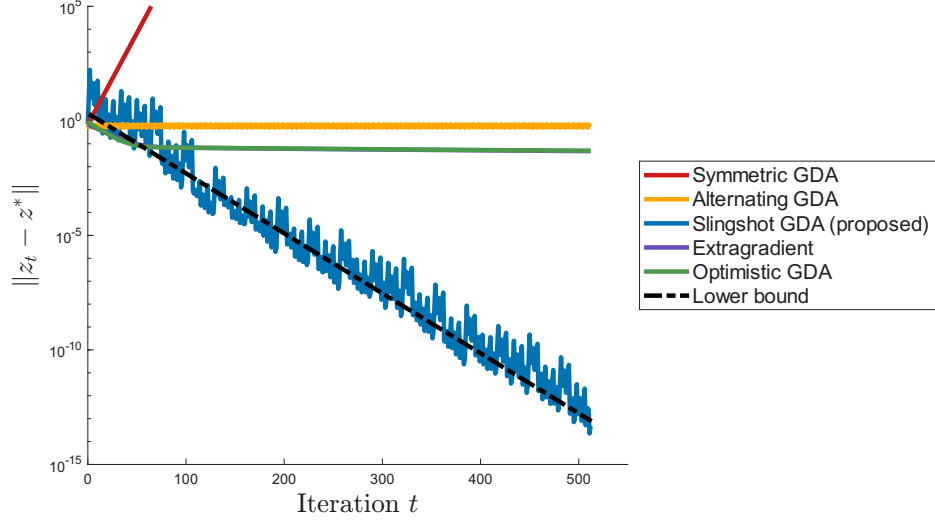


FIGURE 3: Comparison of algorithms on a random bilinear problem. Using standard stepsize schedules, GDA either diverges (red) or cycles (yellow). Known ways of making GDA converge require changing the algorithm and do not always lead to fast convergence (purple, green). The proposed slingshot stepsize schedule requires no changes to GDA beyond stepsizes, and leads to the optimally accelerated convergence rate for bilinear problems. Performance on this random problem is similar to worst-case problems (dotted minimax lower bound); qualitatively similar behavior is observed for other problems. Further details in [Appendix B.5](#).

1.4 Interpretation via second-order approximation

Here we provide a complementary interpretation of the two-step progress made by the slingshot stepsize schedule. This interpretation extends beyond the setting of linear ∇f : it is exact if ∇f is linear, and holds approximately for the general setting where ∇f is nonlinear.

The overarching idea is that positive/negative steps in consecutive iterations cancel to first order, leading to a second-order movement in a new direction that leads to convergence and is otherwise impossible for GDA to move in. For intuition, a useful analogy is *parallel parking*. Parallel parking a car requires paired movements: first move forward (at an angle by turning the wheel), then backward by the same amount (at a slightly different angle by turning the wheel). The car’s forward/backward movement cancel to first order, resulting in a second-order movement in a new direction (sideways) that leads to parking and is otherwise impossible for the car to move in. Geometrically, the alternating use of the two vector fields ∇f and $-\nabla f$ leads to a second-order drift (the new direction) which quantifies the *non-reversibility* of gradient descent/ascent; see §5 for Lie bracket/commutator-type calculations and interpretations. This should be contrasted with standard GDA, which always follows the same vector field $(-\nabla_x f, \nabla_y f)$.

The second-order movement can be explicitly computed by a Taylor expansion. For example, for the nonlinear version (1.6) of the slingshot stepsize schedule, the expected movement from two steps of GDA is:

$$\mathbb{E} \begin{bmatrix} x_{2t+2} \\ y_{2t+2} \end{bmatrix} \approx \begin{bmatrix} x_{2t} \\ y_{2t} \end{bmatrix} + \underbrace{\frac{h}{2} \begin{bmatrix} -\nabla_x f(x_{2t}, y_{2t}) \\ \nabla_y f(x_{2t}, y_{2t}) \end{bmatrix}}_{\text{standard direction: GDA update}} - \underbrace{\frac{h^2}{2} \nabla^2 f(x_{2t}, y_{2t}) \nabla f(x_{2t}, y_{2t})}_{\text{new direction: HGD update (from positive/negative cancellation)}} \quad (1.11)$$

up to error terms of order $\mathcal{O}(h^3)$. See §5 for details. This expansion (1.11) can be interpreted as a combination of two movements: a standard “forward” GDA step and a new direction that is

impossible in a single GDA step. Notice that $h/2$ in the first direction arises from $\mathbb{E}[\alpha_{2t} + \alpha_{2t+1}] = \mathbb{E}[\beta_{2t} + \beta_{2t+1}]$, and $h^2/2$ in the second direction arises from $\mathbb{E}[\alpha_{2t}\beta_{2t+1} + \alpha_{2t+1}\beta_{2t}]$; these are both positive due to properties (iii) and (ii) in the definition of slingshot stepsize schedules in §1.2.

New direction: Hamiltonian gradient descent. This new direction is a gradient descent update on $\min_{(x,y)} \Phi(x,y)$ where $\Phi(x,y) = \frac{1}{2}\|\nabla f(x,y)\|^2$ denotes the Hamiltonian. This generalizes Observation 1.1, which only applies to bilinear objectives f . Three remarks are in order. First, this auxiliary problem $\min_{x,y} \Phi(x,y)$ is helpful for solving the original problem $\min_x \max_y f(x,y)$ since minimizers for the former are in correspondence with stationary solutions for the latter (assuming their existence). Second, note that GDA has access neither to gradients of Φ nor to Hessians of f ; the paired cancellation in the slingshot stepsize schedule enables GDA to approximately move in such directions using only standard query access to ∇f . Third, note that for this Hamiltonian descent direction, x and y *coordinate* to jointly minimize the stationary criteria $\Phi(x,y)$. This in contrast to standard GDA, where x and y *compete* to minimize and maximize f , respectively, which gives rise to the standard intuition for GDA’s cycling behavior as illustrated in Figure 1.

Effective two-step movement: consensus optimization. The expansion (1.11) shows that two steps of GDA with the slingshot stepsize schedule approximately implements one step of consensus optimization, a popular algorithm for training GANs that involves both a standard “forward” GDA step and a step of Hamiltonian GD [59]. This algorithm’s name arises from the interpretation that x and y are, in part, coordinating. An interesting difference is that consensus optimization exactly implements (1.11) for structured objectives f using double backpropagation, whereas our algorithm approximately implements (1.11) via finite-differencing—which applies to general f and requires only black-box queries to ∇f . See §5.2 for further discussion of similarities and differences.

From intuition to convergence. This connection to consensus optimization provides some justification for our counterintuitive algorithm, since consensus optimization is well-documented to perform well empirically even in challenging non-convex landscapes like GANs [59]. However, this connection does not directly imply non-asymptotic³ convergence of the slingshot stepsize schedule. This is both because the connection (1.11) is inexact, and more importantly, because the theoretical convergence properties of consensus optimization are still nascent—in particular although recent works have proved elegant convergence results [2, 9, 51, 59], these require strong assumptions and do not apply to standard nonlinear setups such as smooth convex-concave f or even smooth strongly-convex-strongly-concave f . Therefore, when we prove our convergence results for the setting of smooth convex-concave f in §4, rather than building upon this connection to consensus optimization, we instead directly prove progress for two steps of the slingshot stepsize schedule. (In fact, in the other direction, we are hopeful that the convergence results we show for the slingshot stepsizes may be adaptable to prove analogous convergence results for consensus optimization, which would help justify the strong empirical performance of that algorithm over the past decade.)

1.5 Related work

There is an extensive literature on GDA since it and its variants form the primary workhorse for solving large-scale min-max problems. In addition to the works already cited above, here we

³Asymptotic local convergence (without rates) for the slingshot stepsize schedule can be shown for non-convex-non-concave settings by taking arbitrarily small stepsizes—in which case the approximation (1.11) to consensus optimization becomes arbitrarily precise—and applying asymptotic local convergence guarantees for that algorithm, as in [59, Corollary 8].

contextualize further with the closest lines of work. See [Tables 1 to 3](#) for complexities of variants of GDA for the various standard settings studied in this paper.

Performance criteria. One reason that there is an extensive literature is that there are different ways of measuring the error of an approximate min-max solution $z = (x, y)$. Two standard criteria are approximate stationarity $\|\nabla f(z)\|$ and distance to the solution set $\inf_{\text{solution}} z^* \|z - z^*\|$; our results apply to the former criterion in all settings, and also apply to the second criterion in all settings where such a result is possible⁴. Another criterion is the duality gap $\max_{y'} f(x, y') - \min_{x'} f(x', y)$; however that is typically studied in constrained settings as it is potentially infinite in unconstrained settings (e.g., $\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} xy$). The literature has also studied alternative criteria including variations of approximate stationarity specifically for constrained settings [52, 54] and restricted duality gaps for unconstrained settings [37, 61, 67], and it would be interesting to investigate our stepsizes in these contexts as well.

Best/averaged/last iterate. Another reason that there is an extensive literature is that the performance criterion can be measured at either the best iterate $\operatorname{argmin}_{z \in \{z_1, \dots, z_T\}} \|\nabla f(z_t)\|$, the averaged (a.k.a. ergodic) iterate $\frac{1}{T} \sum_{t=1}^T z_t$, or the last iterate z_T . Early convergence rates were primarily for the best and averaged iterates since this can simplify the analyses. However, the last iterate is typically desired for applications and proving such a result has been a major thrust of the recent literature, see for example [2, 22, 29, 31, 32] and the references within. All of our results apply to the last iterate of GDA.

GDA with different stepsize schedules. As described above (see [Figure 1](#) for an illustration and [§2.2](#) for short proofs), all previous versions of GDA fail to converge on simple problems like $\min_x \max_y xy$. However, two popular stepsize schedules have been proposed which can enable convergence in some new settings (although still not problems like $\min_x \max_y xy$), accelerate convergence in settings where GDA is already known to converge, and also help in practice.

Alternating GDA updates one variable at a time by selecting $\alpha_{2t} > 0, \beta_{2t} = 0$ (descent step in x) and $\alpha_{2t+1} = 0, \beta_{2t+1} > 0$ (ascent step in y) in consecutive iterations. This alternation improves divergence to limit cycling for the problem $\min_x \max_y xy$ (see [Figure 1](#)) and can help empirically in some non-convex-non-concave settings such as training GANs [55]. Alternating GDA can also lead to faster convergence in settings where GDA already converges, namely smooth strongly-convex-strongly-concave problems. Specifically, [48] shows that alternating GDA enables taking larger stepsizes, leading to a rate of $\mathcal{O}(\kappa^{1.5} \log 1/\varepsilon)$, which is faster than the $\Theta(\kappa^2 \log 1/\varepsilon)$ fastest possible rate using symmetric stepsizes $\alpha_t = \beta_t$. [92] shows that alternating GDA achieves a $\mathcal{O}(\kappa \log 1/\varepsilon)$ local convergence rate under the additional assumptions of arbitrarily close initialization and continuous Hessian near optimality. This rate $\mathcal{O}(\kappa \log 1/\varepsilon)$ is the optimal rate among first-order methods [36], and our slingshot stepsizes achieve this rate globally and without any additional such assumptions (see [Table 3](#)), and moreover enable convergence in further settings (see [Tables 1 and 2](#)).

Two-timescale GDA uses constant but asymmetric stepsizes (i.e., $\alpha \neq \beta$). This algorithm was proposed for training Generative Adversarial Networks (GANs) where the discriminator and generator (a.k.a. the x and y variable) train at different speeds in practice [35], and is popular in settings with different structural assumptions on the minimization objective $f(\cdot, y)$ and the maximization objective $f(x, \cdot)$. An influential result of [52] shows that for certain non-convex-concave settings

⁴For example, there exist unconstrained convex-concave min-max problems (such as arbitrarily flat quadratic functions) for which it is impossible for any algorithm to achieve non-trivial rates for minimizing the distance to a solution. Details in [Remark 3.9](#).

	Algorithm	Rate	Reference
GDA	Simultaneous GDA	No convergence	Folklore, e.g., [41]
	Alternating GDA	No convergence	[27]
	Slingshot GDA	$\mathcal{O}(\sqrt{\kappa} \log(1/\varepsilon))$	Theorem 3.3
Additional dynamics	Negative Momentum	$\mathcal{O}(\kappa \log(1/\varepsilon))$	[27]
	Optimistic GDA	$\mathcal{O}(\kappa \log(1/\varepsilon))$	[22]
	Extragradient	$\mathcal{O}(\kappa \log(1/\varepsilon))$	[8]
	Heavy Ball on Hamiltonian	$\mathcal{O}(\sqrt{\kappa} \log(1/\varepsilon))$	[8]

TABLE 1: Complexity of GDA variants for unconstrained *bilinear* problems $\min_x \max_y x^\top B y$. Here $\kappa = \sigma_{\max}^2(B)/\sigma_{\min}^2(B)$ where $\sigma_{\min}$ denotes the smallest non-zero singular value, and ε denotes the target accuracy for distance to a solution (rates are identical for $\|\nabla f\|^2$ up to a logarithm). Both [8] and **Theorem 3.3** exactly achieve the accelerated rate $\mathcal{O}(\sqrt{\kappa} \log 1/\varepsilon)$ which is information-theoretically optimal among arbitrary first-order methods.

	Algorithm	Rate	Reference
GDA	Simultaneous GDA	No convergence	Folklore, e.g., [41]
	Alternating GDA	No convergence	[27]
	Slingshot GDA	$\mathcal{O}(1/\varepsilon)$	Theorem 4.4
Additional dynamics	Extragradient	$\mathcal{O}(1/\varepsilon)$	[29, 31, 80]
	Optimistic GDA	$\mathcal{O}(1/\varepsilon)$	[32]
	Extra anchored gradient and variants	$\mathcal{O}(1/\sqrt{\varepsilon})$	[49, 82, 87]

TABLE 2: Complexity of GDA variants for $\|\nabla f\|^2 \leq \varepsilon$ for unconstrained smooth *convex-concave* problems. Here the smoothness $L = 1$; multiply all results by L^2 for general L . A lower bound of $\Omega(1/\sqrt{\varepsilon})$ is shown in [87].

and certain notions of convergence, two-time scale GDA can converge, namely if $f(\cdot, y)$ is Lipschitz (but potentially non-convex), $f(x, \cdot)$ is concave, f is smooth, and the domain of y is compact. Note that changing the notion of convergence is essential since the iterates of two-timescale GDA diverge exponentially fast from the solution of unconstrained bilinear problems due to the time-invariance and positivity of the stepsize schedule, see §2.2 for a proof. Refined convergence rates for two-timescale GDA in this non-convex-concave setting are an active area of research, see e.g., [50, 54] and the references within. More broadly beyond standard GDA, two-timescale dynamics have been useful in a variety of settings including stochastic optimization [15, 52], mean field games [56], extragradient-type algorithms [18], and more.

Negative parameters in optimization. Momentum is a powerful tool in convex optimization [65, 71], but the direct use of (positive) momentum fails to help GDA converge on simple problems like $\min_x \max_y xy$. An interesting idea of [27] is to use *negative momentum*: this can be helpful empirically for training GANs, has interpretations via optimism [60], and leads to convergence in bilinear [27] and strongly-convex-strongly-concave settings [91], albeit at suboptimal

	Algorithm	Rate	Reference
GDA	Simultaneous GDA	$\mathcal{O}(\kappa^2 \log 1/\varepsilon)$	[51]
	Alternating GDA	$\mathcal{O}(\kappa^{1.5} \log 1/\varepsilon)$	[48]
	Slingshot GDA	$\mathcal{O}(\kappa \log 1/\varepsilon)$	Theorem 4.5
Additional dynamics	Negative Momentum	$\mathcal{O}(\kappa^{1.5} \log 1/\varepsilon)$	[91]
	Extragradient	$\mathcal{O}(\kappa \log 1/\varepsilon)$	[84]
	Optimistic GDA	$\mathcal{O}(\kappa \log 1/\varepsilon)$	[60]

TABLE 3: Complexity of GDA variants for unconstrained L -smooth μ -*strongly-convex-strongly-concave* problems. Here $\kappa := L/\mu$ and ε denotes the target accuracy for distance to optimum (rates are identical for $\|\nabla f\|^2$ up to a logarithm). This is the only setting in the paper for which GDA converges with standard stepsizes; the slingshot stepsizes accelerate convergence to match the information-theoretically optimal rate of $\Omega(\kappa \log 1/\varepsilon)$ from [36, 93].

convergence rates [90], see Tables 1 and 3. Negative momentum bears some intuitive similarities to our slingshot stepsizes in that both involve some sort of backward movement, and is therefore perhaps the closest existing algorithm to ours. However, negative momentum is fundamentally different for many reasons: it changes GDA by adding internal dynamics, it has only been used with positive stepsizes, and its use with symmetric GDA (as in [90, 91]) makes the algorithm symmetric and time-invariant. Moreover, it is an open question whether negative momentum can accelerate in bilinear/strongly-convex-strongly-concave settings or even can converge in convex-concave settings, whereas we show that all of this is possible using slingshot stepsizes.

In standard (non-min-max) optimization, negative parameters have barely been investigated. The only references we are aware of are [21, 79], who allow for (but do not necessarily use) both positive and negative stepsizes in their line search in order to exploit negative curvature directions in non-convex problems. This usage of negative stepsizes is for an entirely different reason than in the present paper, and moreover we note that their line search would *not* use negative stepsizes for convex settings.

First-order methods in min-max optimization. As mentioned above, although the stepsize schedules in alternating GDA and two-timescale GDA can be helpful in certain ways, they do not fix the non-convergence issues of GDA in simple settings like unconstrained bilinear problems. Therefore much of the literature on first-order algorithms for min-max optimization has focused on modifying GDA beyond stepsizes to enable convergence.

The first proposed algorithm for solving general convex-concave min-max problems was the *extragradient method* (EGD) [41]. In each iteration (from z_t to z_{t+1}), EGD performs a step of GDA from the current iterate z_t to create an auxiliary iterate z'_t (look-ahead step), and then uses the gradient at z'_t as the update direction from z_t (movement step). Asymptotic convergence for EGD was first shown by [41], and substantial work since has focused on non-asymptotic rates. For convex-concave objectives, [80] proved a $\mathcal{O}(1/\varepsilon)$ rate for best-iterate convergence in squared gradient norm; a matching last-iterate rate was first shown under the additional assumption of Lipschitz Hessians [29] and was later strengthened to only require standard first-order smoothness assumptions [31]. Matching $\mathcal{O}(1/\varepsilon)$ rates have also been shown for the ergodic iterate for the (restricted) gap function [29, 61, 63]. For strongly-convex-strongly-concave problems, [84] showed an optimal $\mathcal{O}(\kappa \log 1/\varepsilon)$ convergence rate for EGD. Our results match these $\mathcal{O}(1/\varepsilon)$ and $\mathcal{O}(\kappa \log 1/\varepsilon)$ rates by using GDA in its original form, see Tables 2 and 3.

A related algorithm is the *past-extragradient method*, nowadays commonly known as *optimistic GDA* (OGDA) [72]. OGDA performs a look-ahead step, similar to EGD, however this look-ahead step is computed differently by using prior gradient information. OGDA has several interpretations and has been studied through the lens of no-regret algorithms [74, 75], as an approximation of the proximal point method [37], and as a variation on Nesterov-style momentum for convex optimization where “momentum” arises from the difference of consecutive gradients rather than the difference of consecutive iterates [60]. Last-iterate convergence results for squared gradient norm have been shown for OGDA, achieving the optimal rate $\mathcal{O}(\kappa \log 1/\varepsilon)$ for strongly-convex-strongly-concave problems [60] and $\mathcal{O}(1/\varepsilon)$ for smooth convex-concave problems [32]. Our results match these $\mathcal{O}(1/\varepsilon)$ and $\mathcal{O}(\kappa \log 1/\varepsilon)$ rates by using GDA in its original form, see Tables 2 and 3.

Recent work achieves accelerated convergence rates in the smooth convex-concave setting, improving $\mathcal{O}(1/\varepsilon)$ to the optimal rate $\mathcal{O}(1/\sqrt{\varepsilon})$ for squared gradient norm [49, 82, 87, 88]. These accelerated methods primarily rely on combining either EGD or OGDA with anchoring, a mechanism similar to Halpern iterations for root finding, where after each update, the current iterate is moved slightly towards the initial iterate [49, 82, 87]. Alternative accelerated algorithms can also

be derived by applying a certain “H-duality” operation to those algorithms [88].

Performance estimation. Many recent advances in the design and analysis of first-order min-max algorithms have been guided by computer-assisted analyses. This line of work is based on *performance estimation problems* (PEP), pioneered by [24]. The overarching idea is that the search for a proof of an algorithm’s convergence can be recast as a certain semidefinite program. These techniques were originally proposed for convex optimization and recently have been adapted to min-max problems in order to design new algorithms and prove tighter convergence rates for existing algorithms [31, 32, 48, 78, 88, 91, 92]. Our analysis of bilinear and quadratic problems uses only simpler spectral-type arguments, but our analysis for convex-concave problems is inspired by PEP techniques, see §4.3 for a discussion of the similarities and differences.

Min-max problems with structure. Specialized techniques can enable faster convergence rates for problems with additional structure. For example, [8] proposed a specialized algorithm specifically for bilinear problems that achieves the optimal accelerated rate. Interestingly, our slingshot stepsizes enable matching this optimal rate while using GDA in its original form, see Table 1. Faster convergence has also been shown under other structural assumptions, for example smooth strongly-convex-strongly-concave problems where the smoothness and strong convexity/concavity parameters are different for x and y [16, 38, 53, 93], and bilinearly coupled problems of the form $f(x, y) = f_x(x) - f_y(y) + x^\top B y$ [17, 19, 26, 28, 42], and it is an interesting question if the slingshot stepsizes can similarly lead to improved rates under such assumptions, see the discussion in §6.

Stepsize optimization in minimization problems. An orthogonal line of work aims to accelerate gradient descent for (non-min-max) convex optimization by judiciously choosing stepsizes. Classically, in 1953, Young showed that for the special case of convex quadratics, using stepsizes based on the roots of Chebyshev polynomials enables accelerating GD [89]. Starting with [4], a recent line of work has shown that stepsize-based acceleration is possible for convex optimization beyond quadratics, see [14, §1.3] for a recent overview of this extremely active area of research. For the purpose of the present paper, the relevance is that all of those results are for (non-min-max) convex optimization; the present paper shows for the first time that these types of ideas can similarly accelerate GDA in min-max optimization. Specifically, in §3 we show how the positive/negative cancellation in the slingshot stepsize schedule reduces accelerating GDA for min-max problems with linear ∇f , to accelerating GD for convex quadratic minimization. This leads to analogously accelerated rates using stepsizes based on roots of Chebyshev polynomials. It seems potentially plausible that in convex-concave settings, the slingshot stepsize schedule could also be accelerated by combining it with ideas from the silver stepsize schedule [5, 7], which enables accelerating GD to the rate $\mathcal{O}(\varepsilon^{-\log_{1+\sqrt{2}} 2}) \approx \mathcal{O}(\varepsilon^{-0.7864})$ for (non-quadratic) smooth convex optimization.

2 Preliminaries

2.1 Min-max optimization

In this paper we consider unconstrained min-max optimization problems, i.e., problems of the form

$$\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} f(x, y). \quad (2.1)$$

For simplicity of exposition, we consider finite-dimensional spaces $\mathbb{R}^{d_x}$ and $\mathbb{R}^{d_y}$, although our results extend to infinite-dimensional Hilbert space ℓ_2 . We focus on the fundamental setting where f is

L -smooth (i.e., ∇f is L -Lipschitz) and convex-concave (i.e., $f(\cdot, y)$ is convex for any y , and $f(x, \cdot)$ is concave for any x). In the paper we illustrate the slingshot stepsize schedule for several settings (bilinear f , quadratic f , convex-concave f , and strongly-convex-strongly-concave f), and for the convenience of the reader, we recall these standard definitions in the corresponding sections.

Solutions and approximate solutions. A saddle point of (2.1) is a point $z^* = (x^*, y^*)$ satisfying

$$f(x^*, y) \leq f(x^*, y^*) \leq f(x, y^*), \quad \forall x, y.$$

Intuitively, such a point is an equilibrium in the sense that the “ x -player” cannot improve the objective by deviating from the strategy x^* if the y -player does not deviate from y^* , and vice versa.

A related solution concept is a stationary point, which is a point $z^* = (x^*, y^*)$ satisfying

$$\|\nabla f(z^*)\| = 0.$$

In general, the saddle-point condition is more stringent than the stationary condition. However, for unconstrained convex-concave problems, the two notions are equivalent and can be respectively interpreted as global and local characterizations of optimality. Hence we use these terms interchangeably. Throughout we assume that such a solution point exists; this precludes degenerate min-max problems such as $f(x, y) = x$ for which the value of (2.1) is not finite and never achieved.

When numerically computing solutions, as is the goal in this paper, one can only hope to find approximate solutions $z = (x, y)$. Standard notions of approximate solutions are proximity to a stationary point $\|z - z^*\| \leq \varepsilon$ and approximate stationarity $\|\nabla f(z)\| \leq \varepsilon$. Note that the results of the former type imply results of the latter type under smoothness (since $\|\nabla f(z)\| = \|\nabla f(z) - \nabla f(z^*)\| \leq L\|z - z^*\|$), however the latter is possible in settings where the former is not (e.g., see Remark 3.9).

2.2 Failure of GDA with standard stepsizes

We motivate our algorithm by first recalling the folklore explanation for why GDA does not converge. Consider even the simple bilinear min-max problem

$$\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} xy. \quad (2.2)$$

The continuous-time GDA flow

$$\begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \end{bmatrix} = \begin{bmatrix} -\nabla_x f(x(t), y(t)) \\ \nabla_y f(x(t), y(t)) \end{bmatrix} = \begin{bmatrix} -y(t) \\ x(t) \end{bmatrix}$$

fails to converge since $x^2 + y^2$ is a conserved quantity. This results in cycling in circles around the unique saddle-point solution $(x^*, y^*) = (0, 0)$. See the flow lines in Figure 1.

This failure extends to discretizations of these continuous-time dynamics. Specifically, the following lemma shows that for standard types of stepsizes, GDA fails to converge in the sense that its iterates remain bounded away from the unique saddle-point solution $(x^*, y^*) = (0, 0)$. This means that GDA also fails to converge in the sense of approximate stationarity. In fact, “does not converge” becomes “diverges” if the stepsizes are non-zero.

Lemma 2.1 (Failure of GDA with standard stepsizes). *GDA does not converge on the min-max problem (2.2) for stepsize schedules $\alpha_t, \beta_t \in \mathbb{R}$ that fall under any of the following categories:*

- (i) Constant stepsizes: $\alpha_t \equiv \alpha$ and $\beta_t \equiv \beta$ for all iterations t , for some α, β .

(ii) Non-negative stepsizes: $\alpha_t, \beta_t \geq 0$ for all iterations t .

(iii) Symmetric stepsizes: $\alpha_t = \beta_t$ for all iterations t .

Proof. For this problem, the dynamics of GDA simplify to the linear dynamical system

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \mathbf{U}_t \begin{bmatrix} x_t \\ y_t \end{bmatrix} \quad \text{where} \quad \mathbf{U}_t = \begin{bmatrix} 1 & -\alpha_t \\ \beta_t & 1 \end{bmatrix}. \quad (2.3)$$

(i) For constant stepsizes, $\mathbf{U}_t \equiv \mathbf{U}$ for all t . It is a basic fact that such a time-invariant linear system converges from arbitrary initialization if and only if the spectral radius $\rho(\mathbf{U}) < 1$. But $\rho(\mathbf{U}) \geq 1$ since the eigenvalues of $\mathbf{U}$ are $1 \pm i\sqrt{\alpha\beta}$ and $\alpha, \beta \in \mathbb{R}$. Hence GDA is not convergent.

(ii) For non-negative stepsizes, $\det(\mathbf{U}_t) = 1 + \alpha_t\beta_t \geq 1$. Since the spectral norm is lower bounded by the spectral radius, which is in turn lower bounded by the geometric mean of the eigenvalues, we have $\|\prod_{t=1}^T \mathbf{U}_t\| \geq \rho(\prod_{t=1}^T \mathbf{U}_t) \geq \sqrt{\det(\prod_{t=1}^T \mathbf{U}_t)} = \prod_{t=1}^T \sqrt{\det(\mathbf{U}_t)} \geq 1$. Hence GDA is not convergent.

(iii) For symmetric stepsizes, $\det(\mathbf{U}_t) = 1 + \alpha_t^2 \geq 1$. Then argue as in (ii). $\square$

In fact, the failure of GDA with these types of stepsize schedules is *universal*: GDA diverges with stepsizes that are constant, non-negative, and/or symmetric not only for the toy problem (2.2), but in fact for *any* bilinear min-max game. The proof essentially just follows by diagonalizing: GDA fails on each coordinate of a general bilinear min-max problem for the same reason that it fails on the univariate example above; details in [Appendix B.2](#).

Lemma 2.2 (Universal failure of standard GDA). *Lemma 2.1 holds even when (2.2) is replaced by an arbitrary bilinear min-max problem $\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} x^\top \mathbf{B} y$, for $\mathbf{B} \neq \mathbf{0}$.*

2.3 Chebyshev polynomials

Our analysis (only) for min-max problems with linear gradients in §3 makes use of classical facts about Chebyshev polynomials. We briefly recall relevant preliminaries here and refer the reader to standard textbooks such as [58, 76] for further background. Throughout we use only Chebyshev polynomials *of the first kind* and therefore drop that modifier for brevity.

Let $\mathcal{T}_n$ denote the degree- n Chebyshev polynomial over the interval $[-1, 1]$. This can be defined in several equivalent ways, for example 2^{n-1} times the monic degree- n polynomial minimizing $\|p\|_{L^\infty[-1,1]}$, or 2^{n-1} times the monic polynomial $\prod_{t=0}^{n-1} (x - x_t)$ with roots $x_t = \cos(\frac{2t+1}{2n}\pi)$. For a compact interval $[a, b]$, the degree- n Chebyshev polynomial is defined via shifting as

$$\mathcal{T}_n^{[a,b]}(\lambda) := \mathcal{T}_n(\mathcal{L}^{[a,b]}(\lambda)) \quad \text{where} \quad \mathcal{L}^{[a,b]}(\lambda) := \frac{2}{b-a} \left(\lambda - \frac{b+a}{2} \right)$$

denotes the monotone linear map sending $[a, b]$ to $[-1, 1]$.

The relevance of Chebyshev polynomials for convex optimization is that the optimal GD stepsize schedules of length T for optimizing m -strongly convex, M -smooth quadratic functions are (any permutation of) the T inverse roots of $\mathcal{T}_T^{[m,M]}$ [89]. Our results in §3 show how these ideas can also improve GDA for min-max optimization with bilinear or quadratic objectives.

3 Min-max problems with linear gradients

In this section we consider min-max problems $\min_x \max_y f(x, y)$ for which ∇f is linear. By using **time-varying, asymmetric, and alternately negative stepsizes**, we show that the iterates of GDA can

be connected to polynomials of $\nabla^2 f$ (which in this setting is constant), and the **optimal stepsizes of GDA are connected to the roots of certain extremal polynomials**. This allows for a simple, explicit analysis of the proposed slingshot stepsize schedule. See §1.3 for a high-level overview.

3.1 Bilinear objectives

We first consider bilinear problems—a classical counterexample for GDA [41]. Since we assume existence of a saddle point $(\tilde{x}, \tilde{y})$, these problems are of the form

$$\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} (x - \tilde{x})^\top \mathbf{B}(y - \tilde{y}). \quad (3.1)$$

By setting $\nabla f(z^*) = 0$, the saddle points $z^* = (x^*, y^*)$ are characterized by $x^* \in \tilde{x} + \text{Ker}(\mathbf{B}^\top)$ and $y^* \in \tilde{y} + \text{Ker}(\mathbf{B})$. We make the standard assumption that $\mathbf{B}$ is spectrally bounded, namely all non-zero squared singular values of $\mathbf{B}$ lie in some interval $[m, M]$. The spectral upper bound can be interpreted as M -smoothness of the Hamiltonian $\Phi(z) = \frac{1}{2} \|\nabla f(z)\|^2$, or equivalently $\sqrt{M}$ -smoothness of f . The spectral lower bound is equivalent to Φ being m -strongly convex; this can be relaxed, see §3.2 for how the slingshot stepsize schedule still obtains the optimal convergence rate in that setting (i.e., where f is any smooth bilinear function).

For such bilinear problems, we propose the following version of the slingshot stepsize schedule. Notice that the stepsizes are asymmetric, time-varying, and alternately positive or negative.

Definition 3.1 (Slingshot stepsize schedules for bilinear min-max optimization). *For any even number of iterations $2T$ and any spectral bounds $0 < m \leq M < \infty$ on the bilinear coupling, the slingshot stepsize schedule is*

$$\alpha_{2t} = -\beta_{2t} = -\alpha_{2t+1} = \beta_{2t+1} = h_t, \quad t \in \{0, 1, \dots, T-1\},$$

where $\{h_t\}_{t=0}^{T-1}$ are any permutation of $\{r_t^{-1/2}\}_{t=0}^{T-1}$, where

$$r_t := \frac{M+m}{2} + \frac{M-m}{2} \cos\left(\frac{2t+1}{2T}\pi\right), \quad t \in \{0, 1, \dots, T-1\},$$

are the T roots of the Chebyshev polynomial $\mathcal{T}_T^{[m, M]}$.

The permutation invariance arises because the updates of GDA are commutative in this setting, at least in exact arithmetic, see §3.3. The relevance of r_t (and hence the magnitudes of the stepsizes h_t) is that they are the T roots of the solution to the following extremal polynomial problem, which as our analysis below shows, governs the fastest possible convergence rate of GDA. Below let $\mathcal{P}_T$ denote the space of degree- T polynomials p satisfying $p(0) = 1$.

Lemma 3.2 (Extremal polynomial lemma for bilinear min-max problems). *Let $T \in \mathbb{N}$, $0 < m < M < \infty$, and $\kappa := M/m$. Then*

$$\min_{p \in \mathcal{P}_T} \max_{\lambda \in [m, M]} |p(\lambda)|$$

has the unique optimal solution

$$p(\lambda) = \frac{\mathcal{T}_T^{[m, M]}(\lambda)}{\mathcal{T}_T^{[m, M]}(0)} = \prod_{t=0}^{T-1} (1 - \lambda/r_t)$$

and corresponding value

$$R_T := \frac{1}{\mathcal{T}_T^{[m, M]}(0)} = \frac{2(\sqrt{\kappa}+1)^T(\sqrt{\kappa}-1)^T}{(\sqrt{\kappa}+1)^{2T} + (\sqrt{\kappa}-1)^{2T}} \approx \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^T.$$

[Lemma 3.2](#) follows from the extremal definition of Chebyshev polynomials in [§2.3](#), see e.g., [\[25, §2\]](#) or [\[4, §2\]](#) for details. [Lemma 3.2](#) classically arises in convex optimization since the roots r_t are the optimal inverse stepsizes for GD for m -strongly convex, M -smooth quadratic minimization, and R_T is the optimal convergence rate for T steps of any first-order optimization algorithm [\[89\]](#).

We show that the slingshot stepsize schedule in [Definition 3.1](#) makes GDA converge. As described in [§1.3](#), using slingshot stepsizes with constant magnitudes h_t is sufficient for exponentially fast convergence, and using the Chebyshev stepsize magnitudes in [Definition 3.1](#) accelerates the convergence to the optimal rate. For simplicity we state the result using distance as the performance criterion; this extends identically to convergence in gradient norm by either re-doing the same analysis or, more simply, by using the fact that distance and gradient norm are equivalent (up to a multiplicative factor depending only on m and M) and thus switching these criteria does not affect the optimal asymptotic rate $\lim_{T \rightarrow \infty} R_T^{1/T}$.

Theorem 3.3 (GDA converges for bilinear min-max optimization). *Consider any integer T , any dimensions d_x, d_y , any initialization $z_0 = (x_0, y_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, and any bilinear min-max problem [\(3.1\)](#) with spectral bounds $0 < m \leq M < \infty$. Using the slingshot stepsize schedule in [Definition 3.1](#), GDA converges to a saddle point z^* at rate*

$$\|z_{2T} - z^*\| \leq R_T \|z_0 - z^*\|.$$

In particular, $\|z_{2T} - z^*\| \leq \varepsilon$ after $\mathcal{O}(\sqrt{\kappa} \log \frac{\|z_0 - z^*\|}{\varepsilon})$ steps.

Proof. Without loss of generality, after a possible translation we may assume that $\tilde{x} = 0$ and $\tilde{y} = 0$ to simplify notation. We show that the iterates converge to the saddle point $z^* = (x^*, y^*)$ that is closest to initialization; that is, let x^* and y^* be the projections of x_0 and y_0 onto $\text{Ker}(\mathbf{B}^\top)$ and $\text{Ker}(\mathbf{B})$, respectively. For this problem, the GDA iterates form a linear dynamical system

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \mathbf{U}_n \begin{bmatrix} x_n \\ y_n \end{bmatrix} \quad \text{where} \quad \mathbf{U}_n = \begin{bmatrix} \mathbf{I} & -\alpha_n \mathbf{B} \\ \beta_n \mathbf{B}^\top & \mathbf{I} \end{bmatrix}.$$

Because of the pairing of consecutive stepsizes in the slingshot stepsize schedule,

$$\mathbf{U}_{2t+1} \mathbf{U}_{2t} = \begin{bmatrix} \mathbf{I} & h_t \mathbf{B} \\ h_t \mathbf{B}^\top & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} & -h_t \mathbf{B} \\ -h_t \mathbf{B}^\top & \mathbf{I} \end{bmatrix} = \begin{bmatrix} \mathbf{I} - h_t^2 \mathbf{B} \mathbf{B}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{I} - h_t^2 \mathbf{B}^\top \mathbf{B} \end{bmatrix}. \quad (3.2)$$

Thus in particular

$$\begin{bmatrix} x_{2T} \\ y_{2T} \end{bmatrix} = \prod_{t=0}^{T-1} \mathbf{U}_{2t+1} \mathbf{U}_{2t} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} p(\mathbf{B} \mathbf{B}^\top) & \mathbf{0} \\ \mathbf{0} & p(\mathbf{B}^\top \mathbf{B}) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix},$$

where $p(z) := \prod_{t=0}^{T-1} (1 - h_t^2 z) = \prod_{t=0}^{T-1} (1 - z/r_t)$ is the extremal polynomial in [Lemma 3.2](#). Thus

$$\|y_{2T} - y^*\| = \|p(\mathbf{B}^\top \mathbf{B})(y_0 - y^*)\| = \|\mathbf{V} p(\mathbf{\Sigma}^2) \mathbf{V}^\top (y_0 - y^*)\| \leq \|p(\mathbf{\Sigma}^2)\| \|y_0 - y^*\| \leq R_T \|y_0 - y^*\|.$$

Above, the first step is because $p(\mathbf{B}^\top \mathbf{B})y^* = \prod_{t=0}^{T-1} (\mathbf{I} - h_t^2 \mathbf{B}^\top \mathbf{B})y^* = y^*$ since $y^* \in \text{Ker}(\mathbf{B})$. The second step is by an SVD decomposition of $\mathbf{B} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$, where $\mathbf{\Sigma}$ is invertible and $\mathbf{U}, \mathbf{V}$ are orthogonal. The third step is by sub-multiplicativity of the norm and orthogonality of $\mathbf{V}$. The fourth step is by [Lemma 3.2](#) and the assumption that the non-zero singular values of $\mathbf{B}$ lie in $[\sqrt{m}, \sqrt{M}]$, which implies that the non-zero eigenvalues of $\mathbf{\Sigma}^2$ lie in $[m, M]$.

By an identical argument (or symmetry), we also have

$$\|x_{2T} - x^*\| \leq R_T \|x_0 - x^*\|.$$

Summing the squares of the above two displays yields $\|z_{2T} - z^*\|^2 \leq R_T^2 \|z_0 - z^*\|^2$. $\square$

The convergence rate R_T in [Theorem 3.3](#) is exactly optimal not just among all possible GDA stepsize schedules, but also among all first-order algorithms. Matching lower bounds are known for the class of symmetric Krylov-subspace algorithms [\[8, 36\]](#), which are algorithms for which $(x_t, y_t) \in (x_0, y_0) + \text{Span}\{(-\nabla_x f(x_s, y_s), \nabla_y f(x_s, y_s)) : s < t\}$, and as stated below, this extends to asymmetric Krylov-subspace algorithms, which are algorithms for which $x_t \in x_0 + \text{Span}\{\nabla_x f(x_s, y_s) : s < t\}$ and $y_t \in y_0 + \text{Span}\{\nabla_y f(x_s, y_s) : s < t\}$. This generalization allows for asymmetric updates to x and y , and thus captures the slingshot stepsizes (as well as other algorithms). The proof is a straightforward extension of the argument for symmetric algorithms, hence we defer it to [Appendix B.3](#).

Theorem 3.4 (Optimality of [Theorem 3.3](#)). *For any integer T , any initialization $z_0 = (x_0, y_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, and any asymmetric Krylov-subspace algorithm, there exists a bilinear problem [\(3.1\)](#) for which the initialization is not a saddle point and*

$$\|z_{2T} - z^*\| \geq R_T \|z_0 - z^*\|$$

holds for all saddle points $z^* = (x^*, y^*)$.

An immediate corollary of our results is the resolution of an open question about more sophisticated variants of GDA. State-of-the-art complexity results show that extragradient, optimistic, and negative-momentum variants of GDA converge on bilinear problems but only at the unaccelerated rate $\mathcal{O}(\kappa \log 1/\varepsilon)$ [\[22, 27, 51, 90\]](#), and it was open whether they could converge at the accelerated rate $\mathcal{O}(\sqrt{\kappa} \log 1/\varepsilon)$. [Theorem 3.3](#) shows that GDA can achieve this accelerated rate, hence all these algorithms can too—for the trivial reason that GDA is a special case of these more sophisticated algorithms (use the slingshot stepsizes and set the momentum/look-ahead parameters to zero).

The work [\[8\]](#) also achieves the accelerated rate via algorithms that are specially tailored for the unconstrained bilinear setting. Their algorithm effectively implements Polyak’s heavy ball method on the Hamiltonian $\Phi := \frac{1}{2} \|\nabla f\|^2$ (which is a convex quadratic when f is bilinear). Whereas, as explained in [§5.1](#), our algorithm effectively implements GD on the Hamiltonian with the Chebyshev stepsize schedule [\[89\]](#)—an equivalent way of accelerating quadratic minimization. Hence the algorithm of [\[8\]](#) is to the slingshot stepsizes as Polyak’s heavy ball method is to Young’s stepsize.

It is worth remarking that the slingshot stepsizes access the Hamiltonian in an entirely different way from [\[8\]](#), which enables our algorithm to extend beyond bilinear f . The algorithms in [\[8, §5\]](#) rely on computing the operator $F^{\text{real}}(z) := \frac{1}{\eta} [\mathbf{J} \nabla f(z - \eta \mathbf{J} \nabla f(z)) - \mathbf{J} \nabla f(z)]$ where $\mathbf{J} := \mathbb{D}(\mathbf{I}, -\mathbf{I})$. For bilinear f , the diagonal blocks of $\nabla^2 f$ are zero, hence $F^{\text{real}}(z) = -\mathbf{J} \nabla^2 f(z) \mathbf{J} \nabla f(z)$ coincides with $\nabla \Phi = \nabla^2 f(z) f(z)$. However, beyond bilinear f , this is false since $-\mathbf{J} \nabla^2 f(z) \mathbf{J} \nabla f(z) \neq \nabla^2 f(z) \nabla f(z)$. In fact, F^{real} can even be the negative gradient of the Hamiltonian, which is not a direction that encourages stationarity. By taking steps in the direction of the positive/negative gradient $\pm \nabla f$ rather than the monotone operator $\mathbf{J} \nabla f$, the slingshot stepsizes enables approximating $\nabla \Phi$ in general settings beyond bilinear f . See [§5](#) for further discussion.

3.2 Quadratic objectives

We now turn to min-max problems with quadratic objectives. Since we assume the existence of a saddle point $\tilde{z} = (\tilde{x}, \tilde{y})$, without loss of generality these problems are of the form

$$\min_{x \in \mathbb{R}^{d_x}} \max_{y \in \mathbb{R}^{d_y}} \frac{1}{2} \begin{bmatrix} x - \tilde{x} \\ y - \tilde{y} \end{bmatrix}^\top \underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & -\mathbf{C} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} x - \tilde{x} \\ y - \tilde{y} \end{bmatrix} \quad (3.3)$$

We assume that the objective is convex-concave ($\mathbf{A}, \mathbf{C} \geq 0$) and L -smooth ($-L\mathbf{I} \leq \mathbf{H} \leq L\mathbf{I}$). A primary motivation for this setting is that it captures bilinear problems without a spectral lower bound assumption, i.e., it captures arbitrary smooth bilinear problems by considering the special case $\mathbf{A} = 0$ and $\mathbf{C} = 0$. Since the slingshot stepsize schedule obtains optimal convergence rates for the more general setting of quadratic objectives (and not just smooth bilinear objectives⁵), we present the results in this section at that higher level of generality.

For this setting we propose a similar slingshot stepsize schedule.

Definition 3.5 (Slingshot stepsize schedule for quadratic min-max optimization). *For any even number of iterations $2T$ and any L -smooth, convex-concave, quadratic min-max problem, the slingshot stepsize schedule is*

$$\alpha_t = -\beta_t = h_t, \quad t \in \{0, 1, \dots, 2T-1\},$$

where $\{h_t^{-1}\}_{t=0}^{2T-1}$ are any permutation of

$$\rho_t := L \cos\left(\frac{2t+1}{4T+2}\pi\right), \quad t \in \{0, \dots, 2T\} \setminus \{T\},$$

which are the $2T$ non-zero roots of the Chebyshev polynomial $\mathcal{T}_{2T+1}^{[-L, L]}$. Notice that like in [Definition 3.1](#), these roots come in positive/negative pairs because

$$\rho_t = -\rho_{2T-t}.$$

Just like the slingshot stepsize schedule for the bilinear setting in [Definition 3.1](#), the stepsizes here are also asymmetric, time-varying, and paired positive/negative. The difference from the bilinear setting is in the stepsize magnitudes. This arises because the relevant extremal polynomial problem is slightly different in the quadratic setting here, namely the following.

Lemma 3.6 (Extremal polynomial lemma for quadratic min-max problems). *For $T \in \mathbb{N}$ and $L > 0$,*

$$\min_{p \in \mathcal{P}_{2T}} \max_{\lambda \in [-L, L]} |\lambda p(\lambda)|$$

has value $\frac{L}{2T+1}$, achieved by $p(\lambda) = \frac{(-1)^T}{2T+1} \frac{\mathcal{T}_{2T+1}(\lambda/L)}{\lambda/L} = \prod_{t \in \{0, \dots, 2T\} \setminus \{T\}} (1 - \lambda/\rho_t)$.

This lemma follows directly from the equivalent definitions of Chebyshev polynomials in [§2.3](#) and the fact that odd-degree Chebyshev polynomials have a root at zero. Details in [Appendix B.4](#).

We show that these slingshot stepsizes enable GDA to converge for quadratic settings. As in the bilinear setting above, constant stepsize magnitudes h_t are sufficient for convergence, and using the Chebyshev stepsize magnitudes in [Definition 3.5](#) accelerates the convergence to the optimal rate.

Theorem 3.7 (GDA converges for quadratic min-max optimization). *Consider any integer T , any dimensions d_x, d_y , any initialization $z_0 = (x_0, y_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, and any quadratic min-max problem (3.3) that is convex-concave and L -smooth. Using the slingshot stepsize schedule in [Definition 3.5](#), GDA converges at rate*

$$\|\nabla f(z_{2T})\| \leq \frac{L}{2T+1} \|z_0 - z^*\|, \quad (3.4)$$

where z^* is any saddle point. In particular, $\|\nabla f(z_{2T})\| \leq \varepsilon$ after $\mathcal{O}(\frac{L\|z_0 - z^*\|}{\varepsilon})$ steps.

⁵Note that the matching lower bound in [Theorem 3.8](#) is written for quadratic problems, yet the construction is a smooth bilinear problem, which establishes the optimality of the slingshot stepsizes for such problems.

Proof. Without loss of generality, after a possible translation we may assume that $\tilde{x} = 0$ and $\tilde{y} = 0$ to simplify notation. By the paired positive/negative property of the slingshot stepsize schedule, the iterates of GDA satisfy $z_{t+1} = (\mathbf{I} - h_t \mathbf{H}) z_t$. Hence

$$\nabla f(z_{2T}) = \mathbf{H} z_{2T} = \mathbf{H} \prod_{t=0}^{2T-1} (\mathbf{I} - h_t \mathbf{H}) (z_0 - z^*) = \mathbf{H} p(\mathbf{H}) (z_0 - z^*),$$

where $p(\lambda) = \prod_{t=0}^{2T-1} (1 - h_t \lambda) = \prod_{t=0}^{2T-1} (1 - \lambda/\rho_t)$ is the polynomial in Lemma 3.6. By submultiplicativity, diagonalizing, and the extremal property of p in Lemma 3.6, we conclude that

$$\|\nabla f(z_{2T})\| \leq \|\mathbf{H} p(\mathbf{H})\| \cdot \|z_0 - z^*\| \leq \max_{\lambda \in [-L, L]} |\lambda p(\lambda)| \cdot \|z_0 - z^*\| = \frac{L}{2T+1} \|z_0 - z^*\|.$$

□

Note that the convergence rate in Theorem 3.7 holds for any saddle point z^* . The best bound is obtained by taking z^* to be the closest saddle point to the initialization z_0 , namely the projection of z_0 onto the affine subspace $\tilde{z} + \ker(\mathbf{H})$.

The convergence rate in Theorem 3.7 is exactly optimal not just among all possible GDA stepsize schedules, but also among all first-order algorithms. Several other recent works also asymptotically achieve the accelerated convergence rate $\mathcal{O}(1/\varepsilon)$ for $\|\nabla f(z_{2T})\|$ and do so in more general settings. However no prior work has obtained the optimal constants (as we do here), and moreover all prior methods require modifying GDA beyond just stepsizes, e.g., with anchoring [87, 88].

Theorem 3.8 (Optimality of Theorem 3.7 (Theorem 3 of [87])). *For any integer T , and any initialization $z_0 = (x_0, y_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, and any asymmetric Krylov-subspace algorithm, there exists a quadratic min-max problem (3.3) for which the initialization is not a saddle point and*

$$\|\nabla f(z_{2T})\| \geq \frac{L}{2T+1} \|z_0 - z^*\|$$

holds for all saddle points $z^ = (x^*, y^*)$.*

We conclude this section with two remarks.

Remark 3.9 (Alternative performance criteria). *Theorem 3.7 bounds the gradient norm $\|\nabla f(z_{2T})\|$ in terms of the distance $\|z_0 - z^*\|$. In the bilinear setting, results extend to any combination of these performance criteria since distance and gradient norm are equivalent, see the discussion preceding Theorem 3.3. However, such extensions are false for the quadratic setting. Indeed, it is impossible to prove non-trivial rates (i.e., $1 - \delta$ for any $\delta > 0$) for the performance metrics $\frac{\|z_{2T} - z^*\|}{\|z_0 - z^*\|}$ and $\frac{\|z_{2T} - z^*\|}{\|\nabla f(z_0)\|}$ due to pathologically flat functions such as $f(x, y) = \gamma x^2$ at far initializations $z_0 = (x_0, 0)$, where $\gamma > 0$ is sufficiently small. Similar counterexamples preclude non-trivial rates for $\frac{\|\nabla f(z_{2T})\|}{\|\nabla f(z_0)\|}$; consider the same construction, but now with cx added to $f(x)$, for $c \neq 0$.*

Remark 3.10 (Faster convergence under stronger assumptions). *Theorem 3.7 focuses on quadratic min-max problems in the smooth, (non-strongly) convex-concave setting. A line of work [8, 9, 36] considers additional assumptions of strong convexity, strong concavity, and/or refined smoothness bounds on the individual blocks of $\mathbf{H}$, typically considering modifications of GDA using extragredients, momentum, optimism, etc. Such algorithms can sometimes automatically exploit further structure in quadratic objectives to obtain faster convergence. A potential direction for future work is to optimize stepsizes in these various settings in order to leverage additional such assumptions on the blocks of $\mathbf{H}$. Exponentially faster convergence can be obtained, for example, by appropriately adapting the slingshot stepsize schedule under the additional strongly-convex-strongly-concave assumption $\mathbf{A}, \mathbf{C} \geq \mu \mathbf{I}$. See also the future work discussion in §6.*

3.3 Alternative implementations and complex stepsizes

For both the bilinear and quadratic settings, there are many equivalent implementations of the slingshot stepsize schedule that yield the same last iterate and thus the same convergence rate. This is due to invariances of the last iterate of GDA with respect to certain transformations of the stepsize schedule. For example, this includes arbitrary permutations of the stepsize schedule (since the GDA updates commute in the bilinear and quadratic settings). Another example is that for the bilinear setting $\alpha_{2t}, \alpha_{2t+1}, \beta_{2t}, \beta_{2t+1}$ can be arbitrary solutions of the equations

$$\alpha_{2t} + \alpha_{2t+1} = \beta_{2t} + \beta_{2t+1} = 0 \quad \text{and} \quad \alpha_{2t}\beta_{2t+1} = \alpha_{2t+1}\beta_{2t} = h_t^2 > 0, \quad (3.5)$$

since this is what is needed for the 2-step identity (3.2) to hold in our analysis. These relate respectively to properties (iii) and (ii) of the general definition of slingshot stepsize schedules in §1.2. Note that this system of equations (3.5) has infinite solutions, not just $\alpha_{2t} = \beta_{2t+1} = h_t$ and $\beta_{2t} = \alpha_{2t+1} = -h_t$ as in Definition 3.1.

Perhaps the most striking equivalent implementation uses *purely imaginary stepsizes*:

$$\alpha_{2t} = \beta_{2t} = ih_t \quad \text{and} \quad \alpha_{2t+1} = \beta_{2t+1} = -ih_t, \quad \forall t = 0, 1, 2, \dots, T-1. \quad (3.6)$$

Note that this is a solution to (3.5). This use of complex stepsizes is counterintuitive since the problem has purely real data. We are not aware of complex stepsizes being used anywhere else in optimization or min-max optimization. In forthcoming work we investigate complex stepsizes in detail, their connection to complex differentiation, and the interesting fact that this corresponds to extragradient GDA in the appropriate limit as the stepsizes become small. In this paper we focus on the implementation in Definition 3.1 as it extends gracefully beyond bilinear objectives. (For general f , it is not even necessarily true that f admits an analytic continuation.)

Another notable implementation uses random stepsizes, which are i.i.d. among pairs, and within pairs use the structure of the slingshot stepsize schedule. Unlike the aforementioned implementations which are all equivalent to each other, this produces a different rate for any finite number of iterations; however, it achieves the same optimal rate asymptotically; see Appendix A.1 for details.

Of course, all of this discussion of “equivalent formulations” requires exact arithmetic; in Appendix A.2 we isolate a particular implementation that is more numerically stable with respect to inexact updates and arithmetic.

4 Min-max problems with nonlinear gradients

We now turn to min-max problems $\min_x \max_y f(x, y)$ for which $\nabla f(x, y)$ is nonlinear. In this setting, the connection between GDA iterates and polynomials breaks down, necessitating a fundamentally different type of analysis from §3. Nevertheless, we show that the same main results still hold: GDA can be made to converge by using time-varying, asymmetric stepsizes that are sometimes negative. The particular stepsizes we use are the following.

Definition 4.1 (Slingshot stepsize schedule for convex-concave problems). *Independently for each $t = 0, \dots, T-1$, choose the stepsizes in iterations $2t$ and $2t+1$ as follows. With probability $1/2$,*

$$\alpha_{2t} = -\beta_{2t} = \beta_{2t+1} = h \quad \text{and} \quad \alpha_{2t+1} = 0,$$

and otherwise

$$-\alpha_{2t} = \beta_{2t} = \alpha_{2t+1} = h \quad \text{and} \quad \beta_{2t+1} = 0.$$

This version of the slingshot stepsize schedule is largely similar to the version for min-max problems with linear gradients in §3. A difference is that here we use randomness to break the symmetry for whether the x or y variable uses the negative stepsize in each batch of two iterations: $\beta_{2t} < 0$ in the first event, and $\alpha_{2t} < 0$ in the second event. We note that there are many versions of Definition 4.1 that lead to similar analyses and results as in this section. For example, we do not have to use only one negative step per two iterations. However, a key aspect of any convergent stepsize schedule (that is not satisfied by the stepsizes in §3) is the following:

Remark 4.2 (Positive net movement is necessary for convergence in convex-concave problems). *For the stepsizes in Definition 4.1, $\mathbb{E}[\alpha_{2t} + \alpha_{2t+1}] = \mathbb{E}[\beta_{2t} + \beta_{2t+1}] > 0$. If this were zero (as is the case for the optimal stepsizes for linear ∇f in §3 due to paired positive/negative stepsizes of equal magnitude), then GDA would not converge for smooth convex-concave problems due to pathological counterexamples that are locally linear. For example, consider $\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} f(x, y)$ where $f(x, y)$ is the Huber loss function that is $\|x\|^2/2$ for $|x| \leq 1$ and otherwise $|x| - 1/2$. For GDA initializations bounded away from $[-1, 1]$ for the x variable, such schedules cycle since the positive/negative steps exactly cancel and lead to the same iterate in every other iteration.*

The use of negative stepsizes precludes the possibility of proving a “1-step progress lemma” for this algorithm. Nevertheless, following the same principle from the linear setting in §3, here we establish progress after *two* steps.

Lemma 4.3 (2-step progress lemma). *Let $f(x, y)$ be an L -smooth, μ -strongly-convex-strongly-concave⁶ function with saddle point $z^* = (x^*, y^*)$. Then by using the slingshot stepsize schedule in Definition 4.1 with $h \leq \frac{1}{3L}$, the iterates of GDA satisfy*

$$\mathbb{E}[\|z_{2t+2} - z^*\|^2] \leq (1 - h\mu) \mathbb{E}[\|z_{2t} - z^*\|^2] - \frac{h^2(1 - 3Lh)}{2} \mathbb{E}[\|\nabla f(z_{2t})\|^2], \quad (4.1)$$

where the expectation is over the randomness in the stepsize schedule.

This lemma applies to (non-strongly) convex-concave objectives by taking $\mu = 0$. Below, we show how this lemma directly implies polynomial convergence of GDA for convex-concave objectives in §4.1 and exponential convergence for strongly-convex-strongly-concave objectives in §4.2, and then we prove the lemma in §4.3.

4.1 Convex-concave objectives

Here we show that the slingshot stepsizes enable GDA to converge on unconstrained smooth convex-concave problems—a classical counterexample for GDA that generalizes the bilinear setting.

Theorem 4.4 (GDA converges for convex-concave objectives). *Let $f(x, y)$ be an L -smooth, convex-concave function with saddle point $z^* = (x^*, y^*)$. Then for any $T \in \mathbb{N}$, $h < \frac{1}{3L}$, and initialization $z_0 = (x_0, y_0)$, by using the slingshot stepsize schedule in Definition 4.1, the iterates of GDA satisfy*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(z_{2t})\|^2] \leq \frac{2\|z_0 - z^*\|^2}{h^2(1 - 3Lh)T}. \quad (4.2)$$

In particular, using stepsize parameter $h = \frac{c}{L}$ for any constant $c \in (0, \frac{1}{3})$ ensures

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla f(z_{2t})\|^2] \lesssim \frac{L^2\|z_0 - z^*\|^2}{T}.$$

⁶Recall that a function g is said to be μ -strongly convex if $g(u) \geq g(v) + \langle \nabla g(v), u - v \rangle + \frac{\mu}{2}\|u - v\|^2$ for all u, v , and a function $f(x, y)$ is said to be μ -strongly-convex-strongly-concave if $f(\cdot, y)$ and $-f(x, \cdot)$ are μ -strongly convex for every y and x . If f is twice-differentiable (not required by our results), then this amounts to $\nabla_{xx}^2 f, -\nabla_{yy}^2 f \geq \mu I$.

Proof. For $t = 0, \dots, T-1$, applying Lemma 4.3 with $\mu = 0$ gives

$$\frac{h^2(1-3Lh)}{2} \mathbb{E} [\|\nabla f(z_{2t})\|^2] \leq \mathbb{E} [\|z_{2t} - z^*\|^2] - \mathbb{E} [\|z_{2t+2} - z^*\|^2]. \quad (4.3)$$

Summing these inequalities over t , telescoping, and crudely bounding $\mathbb{E} \|z_{2T} - z^*\|^2 \geq 0$ gives

$$\frac{h^2(1-3Lh)}{2} \sum_{t=0}^{T-1} \mathbb{E} [\|\nabla f(z_{2t})\|^2] \leq \mathbb{E} [\|z_0 - z^*\|^2].$$

Re-arranging and dividing by T completes the proof. $\square$

Theorem 4.4 shows that on average, the iterates are nearly stationary. In practice, this can be implemented by choosing a random stopping time τ uniformly at random from $\{0, \dots, T-1\}$ and then outputting the last iterate $f(z_{2\tau})$. This is why we refer to this result as a “last iterate” result. Indeed, Theorem 4.4 then implies that

$$\mathbb{E} [\|\nabla f(z_{2\tau})\|^2] \lesssim \frac{L^2 \|z_0 - z^*\|^2}{T} \quad (4.4)$$

where the expectation is now over the choice of stopping time τ as well as the randomness in the slingshot stepsize schedule. Alternatively, one can output the iterate in the trajectory with minimal $\|\nabla f(z_{2t})\|^2$; this enjoys the same performance guarantee in the worst case and can provide better performance in practice as it is adaptive to the problem.

These $\mathcal{O}(1/T)$ rates match many popular algorithms that modify beyond GDA beyond just stepsizes, e.g., extragradient [31] and optimistic GDA [32]. An accelerated rate of $\mathcal{O}(1/T^2)$ was recently achieved by a clever combination of extragradient and anchoring [87], and it is an interesting question if our results can be similarly accelerated, see the discussion in §6. Figure 4 provides an illustrative numerical comparison of these algorithms.

4.2 Strongly-convex-strongly-concave objectives

In the case of strongly-convex-strongly-concave objectives, GDA converges with constant positive stepsizes (this is the only such setting in the paper), but at a slow rate—namely $\Theta(\kappa^2 \log 1/\varepsilon)$ is the fastest possible rate for symmetric GDA [8], improvable to $\mathcal{O}(\kappa^{1.5} \log 1/\varepsilon)$ using alternating GDA [48].⁷ Here we show that the slingshot stepsize schedule achieves the accelerated rate $\Theta(\kappa \log 1/\varepsilon)$, which matches extragradient and optimistic GDA [61], and moreover is optimal among arbitrary first-order methods without further assumptions on the problem setting [9].

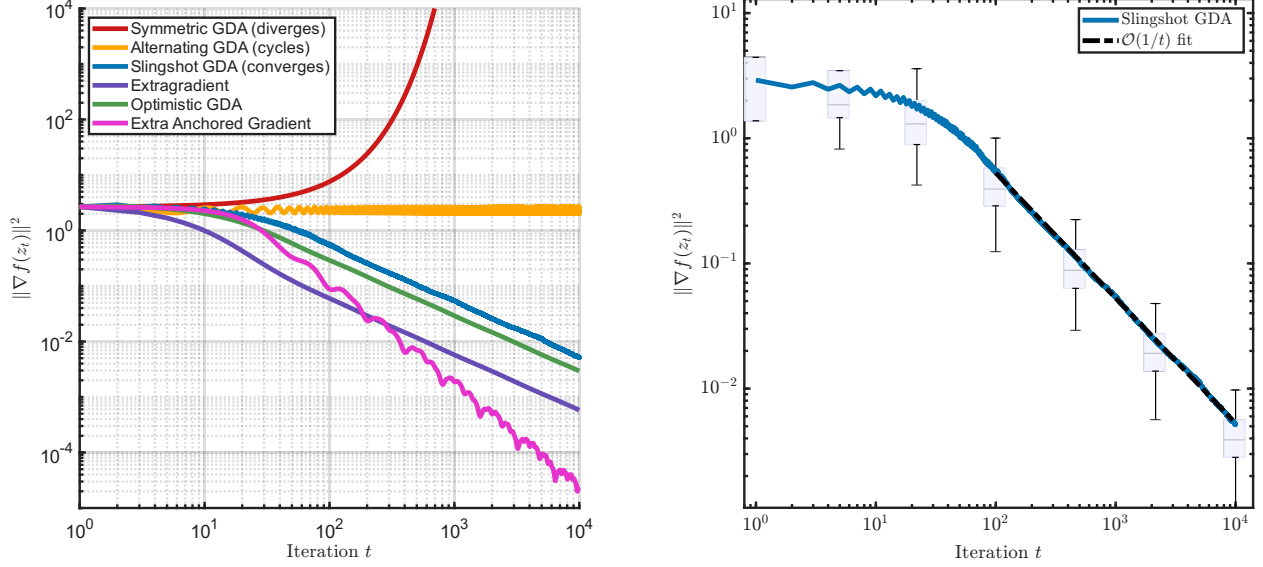
Theorem 4.5 (GDA converges for strongly-convex-strongly-concave objectives). *Let $f(x, y)$ be an L -smooth, μ -strongly-convex-strongly-concave function with saddle point $z^* = (x^*, y^*)$. Then for any $T \in \mathbb{N}$ and initialization $z_0 = (x_0, y_0)$, by using the stepsize schedule in Definition 4.1 with parameter $h \leq \frac{1}{3L}$, the iterates of GDA satisfy*

$$\mathbb{E} [\|z_{2T} - z^*\|^2] \leq (1 - h\mu)^T \|z_0 - z^*\|^2. \quad (4.5)$$

In particular, setting $h = \frac{1}{3L}$, then $\mathbb{E} [\|z_{2T} - z^*\|^2] \leq \varepsilon$ after $\mathcal{O}(\kappa \log \frac{\|z_0 - z^*\|}{\varepsilon})$ steps.

Proof. Applying Lemma 4.3 and dropping the gradient term (as it is non-negative by the assumption on h) yields the contraction $\mathbb{E} [\|z_{2t+2} - z^*\|^2] \leq (1 - h\mu) \mathbb{E} [\|z_{2t} - z^*\|^2]$. Recurse T times. $\square$

⁷GDA with non-negative alternating stepsizes can asymptotically match the accelerated rate $\Theta(\kappa \log 1/\varepsilon)$, but only if one considers local convergence from arbitrarily close initialization and additionally assumes that the Hessian is continuous in this local neighborhood around optimality [92]. Theorem 4.5 shows that the slingshot stepsizes achieve this optimal rate globally and without any such additional assumptions.



(a) Empirical comparison of the convex-concave version of the slingshot stepsize schedule (Definition 4.1).

(b) Variability of slingshot stepsizes across 1000 runs. The $\mathcal{O}(1/\varepsilon)$ empirical rate matches the theory in §4.1.

FIGURE 4: Illustrative numerical comparison on a 1-smooth bilinear problem $\min_{x \in \mathbb{R}^{100}} \max_{y \in \mathbb{R}^{100}} x^\top \mathbf{B} y$ where $\mathbf{B} = \text{diag}(2^0, 2^{-1}, \dots, 2^{-99})$. All algorithms are initialized at the all-ones vector. Regardless of the stepsize choice, symmetric GDA diverges and alternating GDA cycles. Here we run the slingshot stepsize schedule for general convex-concave problems (Definition 4.1); using the version tailored to smooth bilinear problems (Definition 3.5) leads to much faster convergence than all plotted algorithms, with $\|\nabla f(z_t)\|^2 \approx 5 \times 10^{-7}$ at the last iterate $t = 10^4$. We set $h = 2/(9L)$ in Definition 4.1 which minimizes the rate in Theorem 4.4. Extragradient, optimistic GDA, and extra anchored gradient are run with their standard stepsize prescriptions. See Appendix B.5 for further details.

4.3 Proof of 2-step lemma

To simplify notation, without loss of generality let $t = 0$ (so that we measure progress from z_0 to z_2) and let $L = 1$ (by rescaling the objective and stepsize). Recall that the randomness of the proposed stepsize schedule in Definition 4.1 leads to two possible trajectories, each with probability $1/2$. Denote these two trajectories as

$$\begin{aligned} x_1^{(-)} &= x_0 - h \nabla_x f(x_0, y_0), & x_2^{(-)} &= x_1^{(-)}, \\ y_1^{(-)} &= y_0 - h \nabla_y f(x_0, y_0), & y_2^{(-)} &= y_1^{(-)} + h \nabla_y f(x_1^{(-)}, y_1^{(-)}), \end{aligned} \quad (4.6)$$

and

$$\begin{aligned} x_1^{(+)} &= x_0 + h \nabla_x f(x_0, y_0), & x_2^{(+)} &= x_1^{(+)} - h \nabla_x f(x_1^{(+)}, y_1^{(+)}), \\ y_1^{(+)} &= y_0 + h \nabla_y f(x_0, y_0), & y_2^{(+)} &= y_1^{(+)}. \end{aligned} \quad (4.7)$$

We will prove the inequality

$$(1 - h\mu) \|z_0 - z^*\|^2 - c_{h,\mu} \|\nabla f(z_0)\|^2 - \frac{1}{2} \left(\|z_2^{(-)} - z^*\|^2 + \|z_2^{(+)} - z^*\|^2 \right) \geq 0, \quad (4.8)$$

where for shorthand $z_0 := (x_0, y_0)$, $z_2^{(-)} := (x_2^{(-)}, y_2^{(-)})$ and $z_2^{(+)} := (x_2^{(+)}, y_2^{(+)})$, and

$$c_{h,\mu} := \frac{h^2 (1 + (4\mu - 5)h + (6 - 9\mu + 2\mu^2)h^2)}{2(1 + (\mu - 2)h)}.$$

Note that (4.8) implies Lemma 4.3 because for $h < 1/3$, then $c_{h,\mu}$ is increasing in $\mu \in [0, 1]$, and takes the desired minimal value $c_{h,0} = h^2(1 - 3h)/2$ at $\mu = 0$.

Non-negative building blocks. We prove (4.8) by re-writing the left hand side as the sum of three types of non-negative quantities. The simplest such non-negative quantity is based on the assumption that the f is 1-smooth, i.e., ∇f is 1-Lipschitz:

$$P(z, z') := \|z - z'\|^2 - \|\nabla f(z) - \nabla f(z')\|^2 \quad (4.9)$$

is non-negative for any two points $z = (x, y)$ and $z' = (x', y')$. A second type of non-negative quantity that we use is based on the co-coercivity inequality for a convex function g (which will later taken to be $f(\cdot, y)$ for a fixed y , or $-f(x, \cdot)$ for a fixed x):

$$Q_g(v, w) := g(v) - g(w) + \frac{1}{2(1-\mu)} \left[2\langle \nabla g(w) - \mu \nabla g(v), w - v \rangle - \|\nabla g(v) - \nabla g(w)\|^2 - \mu \|v - w\|^2 \right]$$

is non-negative for every pair of points v, w [81]. The final type of non-negative quantity that our proof uses is a sum-of-squares term, with constituent squares

$$\begin{aligned} S_1 &= \frac{1}{2(1-\mu)} \left\| h\mu \nabla_x f(x_0, y_0) - \nabla_x f(x_1^{(+)}, y_1^{(+)}) + \nabla_x f(x^*, y_1^{(+)}) + \mu x_0 - \mu x^* \right\|^2, \\ S_2 &= \frac{1}{2(1-\mu)} \left\| h\mu \nabla_y f(x_0, y_0) - \nabla_y f(x_1^{(-)}, y_1^{(-)}) + \nabla_y f(x_1^{(-)}, y^*) - \mu y_0 + \mu y^* \right\|^2, \\ S_3 &= \frac{1}{2(1-\mu)} \left\| h\mu \nabla_x f(x_0, y_0) + \nabla_x f(x_1^{(-)}, y^*) - \mu x_0 + \mu x^* \right\|^2, \\ S_4 &= \frac{1}{2(1-\mu)} \left\| h\mu \nabla_y f(x_0, y_0) + \nabla_y f(x^*, y_1^{(+)}) + \mu y_0 - \mu y^* \right\|^2, \\ S_5 &= \frac{1}{2\rho(1-\mu)} \left\| \rho \nabla_x f(x_1^{(+)}, y_1^{(+)}) - \nabla_x f(x_1^{(-)}, y_1^{(+)}) - 2h\mu \nabla_x f(x_0, y_0) \right\|^2, \\ S_6 &= \frac{1}{2\rho(1-\mu)} \left\| \rho \nabla_y f(x_1^{(-)}, y_1^{(-)}) - \nabla_y f(x_1^{(-)}, y_1^{(+)}) - 2h\mu \nabla_y f(x_0, y_0) \right\|^2, \\ S_7 &= \frac{1}{2\rho(\rho-h)} \left\| (\rho-h) \nabla f(x_1^{(-)}, y_1^{(+)}) - (\rho+2h(h-1)) \nabla f(x_0, y_0) \right\|^2, \end{aligned}$$

where we denote $\rho := 1 + h\mu - h$.

Certificate of non-negativity. Specifically, the left hand side of (4.8) is equal to h times

$$\begin{aligned} &Q_\phi(x_1^{(+)}, x_1^{(-)}) + Q_\phi(x^*, x_1^{(+)}) + Q_{\phi^*}(x_1^{(-)}, x^*) + Q_\psi(y_1^{(-)}, y_1^{(+)}) + Q_\psi(y^*, y_1^{(-)}) + Q_{\psi^*}(y_1^{(+)}, y^*) \\ &+ \frac{1}{2}P(z_0, z') + \sum_{i=1}^7 S_i, \end{aligned} \quad (4.10)$$

where we denote $z' := (x_1^{(-)}, y_1^{(+)})$, $\phi := f(\cdot, y_1^{(+)})$, $\phi^* = f(\cdot, y^*)$, $\psi := -f(x_1^{(-)}, \cdot)$, and $\psi^* := -f(x^*, \cdot)$.

The fact that the left hand side of (4.8) is equal to h times (4.10) can be checked in a conceptually simple (albeit tedious) manner by expanding out the aforementioned definitions for the three types of terms and plugging in the definitions (4.6) and (4.7) of the algorithm's iterates. For brevity and convenience, we provide a short Mathematica script that rigorously verifies this identity at the URL given in the references [1].

It follows that (4.8) is non-negative, as desired, since (4.10) is the sum of non-negative terms. Note that here we are using the assumption that f is convex-concave because that implies $\phi, \psi, \phi^*, \psi^*$ are convex functions, hence the co-coercivities in (4.10) are non-negative.

Remarks. We make two remarks about this proof. First, the expansion (4.10) enjoys a certain duality: appearances of $x_t^{(-)}$ and $x_t^{(+)}$ are respectively complemented by appearances of $y_t^{(+)}$ and $y_t^{(-)}$. This occurs in the co-coercivities (note the symmetry between the first three terms and the second three terms), the smoothness term (note the duality in $z' = (x_1^{(-)}, y_1^{(+)})$), and in the sum-of-squares terms (note that S_1, S_2 and S_3, S_4 and S_5, S_6 are paired in this way; S_7 is the combination of two such terms as it involves the full-vector gradient).

Second, this analysis approach—expressing the desired quantity in terms of non-negative quantities involving co-coercivities, smoothness inequalities, and sum-of-squares terms—is motivated by the Performance Estimation Problem (PEP) framework pioneered by [24]. PEP has been recently used in many min-max settings [77, 78]. However, typical PEP approaches rely on (variations of the) elegant fact from (non-min-max) convex optimization that in order to prove the tightest possible convergence rates for a first-order algorithm, it is sufficient to only use function and gradient evaluations at the algorithm’s iterates [81]. In contrast, our analysis above—and any alternative analysis—must also incorporate function and gradient evaluations that are *not* at the algorithm’s iterates. For example, function and gradient evaluations at $(x_1^{(-)}, y_1^{(+)})$ appear in the sum-of-squares terms, the co-coercivity terms, and the smoothness term.

5 Connections to consensus optimization and finite differencing

Here we interpret the slingshot stepsize schedule through the second-order expansion of its dynamics. This leads to connections to finite-differencing techniques as well as to Hamiltonian gradient descent and the consensus optimization algorithm. In order to focus on the main conceptual ideas, the derivations in this section are kept informal and we assume for simplicity that $f \in \mathcal{C}^2$. (Note that none of our results in §4 require this assumption; in particular, our results only require the standard assumption that ∇f is Lipschitz continuous.) For concreteness and simplicity, we focus on the implementations of the slingshot stepsize schedule in §3 and §4; see Appendix B.1 for the general case. See also §1.4 for a high-level overview of the derivations in this section.

5.1 Linear version of the slingshot stepsize schedule

Second-order expansion. Recall from §3 that for problems with linear gradients ∇f , we proposed versions of the slingshot stepsize schedule that use paired positive/negative steps of equal magnitude (see Definition 3.1 and Definition 3.5 for the settings of bilinear and quadratic objectives, respectively). By concatenating $z_t = (x_t, y_t)$, this amounts to a two-step update which alternates moving in a positive and negative gradient direction of f :

$$\begin{aligned} z_{2t+1} &= z_{2t} + h\nabla f(z_{2t}) \\ z_{2t+2} &= z_{2t+1} - h\nabla f(z_{2t+1}). \end{aligned} \tag{5.1}$$

Although this version of the slingshot stepsize schedule is designed specifically for the setting of linear ∇f , it is insightful to consider the dynamics for general objectives with nonlinear ∇f . Specifically, below we expand the two-step update from z_{2t} to z_{2t+2} for general nonlinear ∇f . (For linear ∇f , as in §3, all the following approximations are equalities.) A Taylor approximation gives

$$\nabla f(z_{2t+1}) = \nabla f(z_{2t} + h\nabla f(z_{2t})) \approx \nabla f(z_{2t}) + h\nabla^2 f(z_{2t})\nabla f(z_{2t}) \tag{5.2}$$

up to lower-order error terms. Plugging this into (5.1) yields

$$z_{2t+2} = z_{2t} + h\nabla f(z_{2t}) - h\nabla f(z_{2t+1}) \approx z_{2t} - h^2\nabla^2 f(z_{2t})\nabla f(z_{2t}) \tag{5.3}$$

up to lower-order error terms. Note that this expansion (5.3) is invariant with respect to the order of the positive/negative steps in (5.1).

Geometric interpretation. The second-order expansion (5.3) quantifies the *non-reversibility* of gradient descent: performing a step of gradient descent followed by a step of gradient ascent, both of the same stepsize h , does not return to the starting point z . Instead, there is a second-order drift $-h^2 \nabla^2 f(z) \nabla f(z)$. This drift's direction $\nabla^2 f(z) \nabla f(z)$ is the directional derivative of the gradient along itself (described more below) and can be interpreted as the *acceleration* of gradient ascent or gradient descent; this is perhaps most easily seen in continuous-time:

$$\frac{d}{dt} z(t) = \pm \nabla f(z(t)) \quad \implies \quad \frac{d^2}{dt^2} z(t) = \nabla^2 f(z(t)) \nabla f(z(t)).$$

Geometrically, the invariance of the second-order expansion (5.3) with respect to the order of the ascent and descent step amounts to the fact that the Lie bracket of the vector fields ∇f and $-\nabla f$ is zero, i.e., the commutator $[\nabla f, -\nabla f]$ vanishes.

Connection to HGD. The second order expansion (5.3) of this two-step update is equivalent to one step of Hamiltonian gradient descent (HGD). HGD is an algorithm used for solving multiplayer games [2, 10, 59] and amounts to taking gradient descent steps in the concatenated variable $z = (x, y)$ with respect to the Hamiltonian function $\Phi(z) = \frac{1}{2} \|\nabla f(z)\|^2$, the idea being that moving in a descent direction of the Hamiltonian encourages stationarity. Note that both variables minimize the same potential, effectively colluding rather than competing as in the original min-max problem. The update of HGD is given by

$$z_{t+1} = z_t - \gamma \nabla \Phi(z_t) = z_t - \gamma \nabla^2 f(z_t) \nabla f(z_t), \quad (5.4)$$

where the second equality is by the chain rule. Observe that this HGD update (5.4) exactly matches the second-order approximation (5.3) of the slingshot stepsize schedule.

HGD is known to converge quickly in some settings, for example on functions whose Hamiltonians are smooth and satisfy a Polyak-Lojasiewicz condition [2]. However, these conditions on the Hamiltonian do not translate to standard assumptions on f ; indeed HGD fails to converge even on simple objectives that are smooth and convex-concave, such as the 1-dimensional counterexample in Remark 4.2: consider the Huber function $f(x, y)$ that is $x^2/2$ for $|x| \leq 1$ and otherwise is $|x| - 1/2$. If HGD is initialized outside of $[-1, 1]$, then $\nabla \Phi = 0$, hence HGD updates never move.

Connection to finite differencing. Note that the GDA algorithm uses only first-order information, yet effectively yields a two-step update (5.3) that incorporates second-order information. This use of higher-order information is made possible because the slingshot stepsize schedule enables GDA to implicitly implement finite differencing. In particular, here the relevant second-order information is the directional derivative $D_{\nabla f(z)/\|\nabla f(z)\|} \nabla f(z)$ of the function $\nabla f(\cdot)$ in the direction $\nabla f(z)/\|\nabla f(z)\|$, evaluated at z . That is,

$$D_{\nabla f(z)/\|\nabla f(z)\|} \nabla f(z) = \nabla^2 f(z) \frac{\nabla f(z)}{\|\nabla f(z)\|} \approx \frac{\nabla f(z + h \nabla f(z)) - \nabla f(z)}{h \|\nabla f(z)\|}. \quad (5.5)$$

Indeed, this is precisely the key Taylor expansion in (5.2), modulo rearranging and rescaling.

Conceptually, this use of finite differencing is similar to how zeroth-order optimization algorithms compute directional derivatives of a function f using finite-difference approximations [20, 68].

A subtle but important distinction is that in zeroth-order methods, the finite-differencing approximation $\nabla f(z) \approx \frac{\nabla f(z+u) - \nabla f(z)}{\|u\|}$ uses two evaluations within a small distance $\|u\| \ll 1$, which leads to simple bounds for the approximation [68]. In contrast, the slingshot stepsize schedule approximates the directional derivative of $\nabla f(z)$ by using two evaluations within distance $h\|\nabla f(z)\|$, and unless it is already known that the algorithm is converging, this quantity is not actually known to be small. This necessitates different types of analyses than is standard in the finite-differencing literature.

5.2 Nonlinear version of the slingshot stepsize schedule

Second-order expansion. We now provide the analogous expansion for two steps of GDA using the slingshot stepsize schedule for nonlinear ∇f in Definition 4.1. Recall from §4 that this schedule uses randomization to break the symmetry between whether x or y uses a negative stepsize. Following the notation in §4.3, let us distinguish these two trajectories using superscripts $(+)$ and $(-)$. The first step of GDA in these two trajectories is $z_{2t+1}^{(+)} = z_{2t} + h\nabla f(z_{2t})$ and $z_{2t+1}^{(-)} = z_{2t} - h\nabla f(z_{2t})$. Identical to (5.2), a Taylor approximation of ∇f in both cases yields

$$\nabla f\left(z_{2t+1}^{(+)}\right) \approx \nabla f(z_{2t}) + h\nabla^2 f(z_{2t})\nabla f(z_{2t}), \quad \nabla f\left(z_{2t+1}^{(-)}\right) \approx \nabla f(z_{2t}) - h\nabla^2 f(z_{2t})\nabla f(z_{2t}). \quad (5.6)$$

Plugging this gradient approximation into the second step's update gives the two-step expansion:

$$\begin{aligned} x_{2t+2}^{(+)} &\approx x_{2t} - h^2 \left[\nabla^2 f(z_{2t}) \nabla f(z_{2t}) \right]_x, & x_{2t+2}^{(-)} &= x_{2t} - h \nabla_x f(z_{2t}), \\ y_{2t+2}^{(+)} &= y_{2t} + h \nabla_y f(z_{2t}), & y_{2t+2}^{(-)} &\approx y_{2t} - h^2 \left[\nabla^2 f(z_{2t}) \nabla f(z_{2t}) \right]_y. \end{aligned} \quad (5.7)$$

Above, for a vector $v := (v_x, v_y) \in \mathbb{R}^{d_x \times d_y}$, we write $[v]_x$ to denote v_x , and similarly for the y component. Observe from this two-step expansion (5.7) that in the $(+)$ trajectory, the x variable effectively performs one step of HGD, and the y variable performs one step of gradient ascent; and vice versa in the $(-)$ trajectory. As a consequence, the *expected* two-step update is

$$\mathbb{E}[z_{2t+2}] = \frac{z_{2t+2}^{(+)} + z_{2t+2}^{(-)}}{2} \approx z_{2t} - \frac{h}{2} \mathbf{J} \nabla f(z_{2t}) - \frac{h^2}{2} \nabla^2 f(z_{2t}) \nabla f(z_{2t}) \quad (5.8)$$

up to lower-order error terms, where $\mathbf{J} := \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & -\mathbf{I} \end{bmatrix}$.

The key point is that this second-order expansion (5.8) can be viewed as a convex combination of two distinct updates:

- HGD update: $z_{2t} - h^2 \nabla^2 f(z_{2t}) \nabla f(z_{2t})$.
- GDA update: $z_{2t} - h \mathbf{J} \nabla f(z_{2t})$. (Note that this GDA update uses conventional stepsizes: they are positive and symmetric for x and y).

It is crucial that the update (5.8) incorporates both GDA and HGD, since by themselves, both algorithms fail to converge even on simple unconstrained min-max problems (see §2.2 and §5.1 for simple 1-dimensional counterexamples where GDA and HGD fail, respectively).

Connection to consensus optimization. Interestingly, the second-order expansion (5.8) of this two-step update is equivalent to the consensus optimization algorithm of [59], which updates using a combination of a GDA and HGD step, namely

$$z_{t+1} = z_t - h \mathbf{J} \nabla f(z_t) - \gamma \nabla \Phi(z_t). \quad (5.9)$$

This algorithm has received significant attention in the machine learning community due to its success for training Generative Adversarial Networks (GANs), which is a notoriously challenging setting for min-max algorithms, and in particular is a setting where both GDA and HGD experience trouble converging by themselves [59].

Although elegant convergence results have been proven for consensus optimization, this area is still nascent. Existing results either show non-explicit asymptotic convergence that requires arbitrarily close initialization and growth conditions around optimality [59], or provide non-asymptotic convergence bounds but only apply to bilinear f [51] (and moreover with unaccelerated rates) or require non-standard smoothness and/or Polyak-Lojasiewicz-type assumptions on the Hamiltonian which do not apply for smooth convex-concave f (or even smooth strongly-convex-strongly-concave f) [2, 9]. For example, $f(x, y) = \log \cosh x$ is convex-concave, yet its Hamiltonian $\Phi(x, y) = \frac{1}{2} \|\nabla f(x, y)\|^2 = \frac{1}{2} \tanh^2(x)$ does not satisfy the Polyak-Lojasiewicz condition required by [2]. Therefore, when we prove our convergence results for the setting of smooth convex-concave f in §4, rather than building upon this connection to consensus optimization, we instead directly prove progress for two steps of the slingshot stepsizes. (In the other direction, we are hopeful that our convergence results and techniques may be helpful for analyzing consensus optimization.)

Connections to finite differencing. Just like the linear version of the slingshot stepsize schedule (see §5.1), the nonlinear version here implicitly implements finite-differencing in every pair of steps, enabling GDA to effectively move in directions involving second-order information about the objective, despite only ever using first-order information in its updates. See (5.8).

This way of computing second-order information using finite differencing is different from how consensus optimization and HGD are implemented. Even though both those algorithms require updates involving second-order information $\nabla^2 f(z)$, in practice they are implemented using double backpropagation. In particular, because those algorithms only require second-order information through Hessian-vector products of the form $\nabla^2 f(z) \nabla f(z)$, these quantities can be efficiently computed by differentiating the function $\nabla f(z)$ in the direction $\nabla f(z)$, see [70]. To summarize, consensus optimization implements (5.9) exactly but for objectives with specific structure (f must be the composition of “simple” building blocks whose first/second derivatives are known); whereas the slingshot stepsize schedule approximately implements this update but applies to arbitrary f and only requires black-box access to first-order queries ∇f .

6 Discussion

This paper challenges the conventional wisdom that GDA cannot converge in its original form. Our key algorithmic insight for making GDA converge is the use of time-varying, asymmetric, and sometimes negative stepsizes. This deviates from standard intuition from optimization theory (since negative stepsizes amount to backward movement) as well as monotone operator theory (since asymmetric steps treat the minimization and maximization variables separately). These new conceptual ideas lead to several natural directions for future research, such as the following.

Algorithmic building blocks. The use of time-varying, asymmetric, negative stepsizes not only improves GDA in its original form, but also in some settings can accelerate more sophisticated variants of GDA (with momentum, optimism, extragredients, etc.) for the trivial reason that vanilla GDA is a special case; see §3 for concrete examples where this improves prior complexity bounds. Can these new algorithmic ideas be more seamlessly integrated? With these (or other) algorithmic building blocks? With adaptive methods? More broadly, can other algorithms similarly benefit

from time-varying parameters, asymmetric updates, and/or negative movement (in a way that is similar to or different from negative momentum [27])? In practice or in theory? These opportunities were previously overlooked, and we are hopeful that they may become useful building blocks for the design and analysis of min-max optimization algorithms.

Convergence in more general settings. This paper investigates the settings of unconstrained bilinear, quadratic, and (strongly) convex-concave objectives since these are classical counterexamples for GDA and are standard testing grounds for new algorithms. To what extent do the phenomena in this paper extend to more general settings such as nonconvex-concave objectives, constrained domains, and/or stochastic gradients? These questions are motivated by applications such as games, controls, and deep learning. Can GDA still converge, and if so at what rate? Are the optimal stepsizes qualitatively similar to (any variation of) the slingshot stepsize schedule?

Faster convergence in restricted settings. Is faster convergence possible if the objective function is more structured? One can always decompose $f(x, y) = f_{x,y}(x, y) + f_x(x) - f_y(y)$, and in some applications the “coupled” term $f_{x,y}$ and/or the “uncoupled” terms f_x, f_y have additional structure. One potential direction is if the coupled term $f_{x,y}(x, y) = x^\top \mathbf{B}y$ is bilinear, which is motivated for example by Fenchel games [86] and Lagrangian dualization of constrained minimization problems [13]; recent work has designed faster algorithm for example when $\mathbf{B}$ is square and full rank [42]. A second potential direction is if the uncoupled terms f_x and f_y are strongly convex and smooth with different parameters; refined accelerated rates are known for other first-order algorithms [16, 38, 53, 93], and it is natural to ask if these can also be achieved by GDA. A third potential direction is if the Hessians of f_x and f_y have additional spectral structure; such structure has been exploited in (non-min-max) convex optimization [23, 33, 40, 69]. A fourth potential direction is when f_x and f_y admit computationally tractable proximal operators, which is motivated for example by applications in image reconstruction and signal processing [19, 26, 28].

Robustness. This paper shows that interspersing negative stepsizes—occasionally of very large magnitude—helps GDA converge. We expect that such stepsizes may make the trajectory of GDA more sensitive to rounding errors, incorrect modeling assumptions, and imprecise or stochastic gradient calculations. Indeed, the motivation of the slingshot stepsize schedule is that positive/negative steps of size h cancel to provide a net movement of lower-order size $\Theta(h^2)$ in the direction of convergence, see §5, and this delicate cancellation of higher-order movement could be impacted by modeling error. As a first step towards this, in Appendix A.2 we describe a numerically stable fractal-like ordering of the slingshot stepsize schedule for the bilinear setting, by leveraging the connection to convex quadratic optimization in Observation 1.1 and then appealing to known stability ideas in that context. It would be interesting to more broadly understand the tradeoff between convergence rate and stability. As an illustrative example, for unconstrained bilinear problems in the setting of inexact gradients, what is the Pareto-optimal frontier of convergence vs stability—among all GDA stepsize schedules, or more generally among all first-order algorithms?

Randomization and derandomization. In all the different settings, the slingshot stepsize schedule breaks symmetry between the minimization variable x and the maximization variable y . Within each pair of iterations, one must choose which variable to update with a positive step and which to update with a negative (or zero) step. For min-max problems with linear gradients, this intra-pair order does not affect convergence (assuming exact arithmetic), see §3.3. For problems with nonlinear gradients, we break symmetry by randomizing the order between x and y within

every pair of iterations, as this simplifies the design and analysis of the slingshot stepsize schedule, see §4. We conjecture that this result can be de-randomized. Note that this randomization for breaking *intra*-pair symmetry of stepsize signs is distinct from the randomization investigated in Appendix A.1 for breaking *inter*-pair symmetry of stepsize magnitudes.

Implicit bias. The proof of Theorem 3.3 not only shows that GDA converges to a saddle point, but moreover isolates which one: it is the closest solution to initialization, namely x^* and y^* are the respective projections of x_0 and y_0 onto the solution sets $\tilde{x} + \text{Ker}(\mathbf{B}^\top)$ and $\tilde{y} + \text{Ker}(\mathbf{B})$. This can be interpreted as a form of “implicit bias” for GDA, and mirrors the well-known fact that GD always converges to the closest solution for convex quadratic minimization. It would be interesting to understand the implicit bias of the slingshot stepsize schedule beyond the bilinear setting.

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A Alternative implementations

Here we illustrate how the slingshot stepsize schedule can be augmented with other ideas from the literature on stepsize schedules for (non-min-max) convex optimization. We consider i.i.d. random stepsizes in [Appendix A.1](#) and stable fractal orderings in [Appendix A.2](#). For simplicity we focus here on the bilinear setting; it seems plausible that similar ideas may extend to more general settings.

A.1 Random stepsizes

Here we show how to obtain the same optimal convergence rate in [Theorem 3.3](#) by using *random* stepsizes. This is inspired by the recent use of i.i.d. stepsizes in convex optimization [\[4, 6, 39, 73, 94\]](#). Like the deterministic stepsizes in [Definition 3.1](#), these random stepsizes circumvent the classical GDA counterexamples because they are time-varying, asymmetric, and sometimes negative. One possible benefit of these randomized stepsizes is that they do not require prior knowledge of the number of iterations, and another possible benefit is that the variability in random runs could potentially be exploited for [faster parallelized optimization](#) [\[6, §5.3\]](#).

The proposed random stepsizes are linked in pairs in the same way as the deterministic stepsizes in the same way as in [Definition 3.1](#), i.e.,

$$\alpha_{2t} = -\alpha_{2t+1} = -\beta_{2t} = \beta_{2t+1} = h_t, \quad \text{for } t = 0, 1, \dots, \quad (\text{A.1})$$

but now $h_t^{-1/2}$ are not chosen deterministically (as roots of a Chebyshev polynomial), but instead are i.i.d. from the Arcsine distribution on the interval (m, M) , i.e., the distribution with density

$$\frac{d\mu}{dz} = \frac{1}{\pi\sqrt{(M-z)(z-m)}} \cdot \mathbb{1}[z \in (m, M)]. \quad (\text{A.2})$$

Theorem A.1 (Optimal random stepsizes). *Consider any dimensions d_x, d_y , any initialization $z_0 = (x_0, y_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, and any bilinear min-max problem [\(3.1\)](#) with spectral bounds $0 < m \leq M < \infty$. Using the aforementioned random stepsizes, GDA converges to a saddle point z^* at rate*

$$\|z_{2T} - z^*\| \leq C_T \|z_0 - z^*\| \quad (\text{A.3})$$

where C_T satisfies

$$\lim_{T \rightarrow \infty} (C_T)^{1/T} \stackrel{\text{a.s.}}{=} \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

Proof. By an identical calculation as in our result for deterministic stepsize schedules ([Theorem 3.3](#)), the positive/negative pairing [\(A.1\)](#) of the stepsizes ensures that for every T ,

$$\begin{bmatrix} x_{2T} \\ y_{2T} \end{bmatrix} = \begin{bmatrix} p_T(\mathbf{B}\mathbf{B}^\top) & \mathbf{0} \\ \mathbf{0} & p_T(\mathbf{B}^\top\mathbf{B}) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix},$$

where $p_T(z) := \prod_{t=0}^{T-1} (1 - h_t^2 z) = \prod_{t=0}^{T-1} (1 - z/r_t)$ and $r_t := h_t^{-1/2}$ are i.i.d. from the Arcsine distribution. By the same diagonalization argument as done there, $\|x_{2T} - x^*\| \leq C_T \|x_0 - x^*\|$ and $\|y_{2T} - y^*\| \leq C_T \|y_0 - y^*\|$ where $C_T := \sup_{\lambda \in [m, M]} |p_T(\lambda)|$. Now for any $\lambda \in [m, M]$,

$$|p_T(\lambda)|^{1/T} = \prod_{t=0}^{T-1} |1 - \lambda/r_t|^{1/T} = \exp\left(\frac{1}{T} \sum_{t=0}^{T-1} \log |1 - \lambda/r_t|\right) \xrightarrow{\text{a.s.}} \exp(\mathbb{E}_r \log |1 - \lambda/r|) = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

Above the first step is by the definition of p_T , the second step is by taking an exponential and logarithm, the third step is by the Law of Large Numbers, and the final step is by [the equalizing property of the Arcsine distribution](#) [\[6, Lemma 2.2\]](#). The claim follows. $\square$

We conclude with several remarks about these random stepsizes. 1) Optimality of the above asymptotic rate follows immediately from taking the limit as $T \rightarrow \infty$ of the optimal rate $R_T^{1/T}$ for every finite T , see [Lemma 3.2](#). 2) This random stepsize schedule achieves the optimal convergence rate asymptotically, but not for every finite number of iterations T . One can prove high probability bounds on this convergence rate for finite T via (non-asymptotic) concentration inequalities rather than using the (asymptotic) Law of Large Numbers, see [\[6, §5\]](#). 3) The same permutation invariances apply as in the deterministic case, c.f. [§3.3](#). 4) These random stepsizes appear to be more sensitive to numerical precision and modeling error than the deterministic stepsize schedule proposed in the following section, and therefore we suggest using the latter in practice.

A.2 Stable fractal stepsizes

As discussed in [§3.3](#), the convergence rate of GDA for bilinear problems is invariant with respect to several aspects (e.g., permutations) of the slingshot stepsize schedule—assuming exact arithmetic. However, it is a well-known phenomenon that two algorithms which are identical in exact arithmetic can perform quite differently when implemented numerically. In this spirit, [Figure 5](#) illustrates how rounding errors (even when using unusually high levels of precision) can make some permutations of the slingshot stepsize schedule extremely unstable.

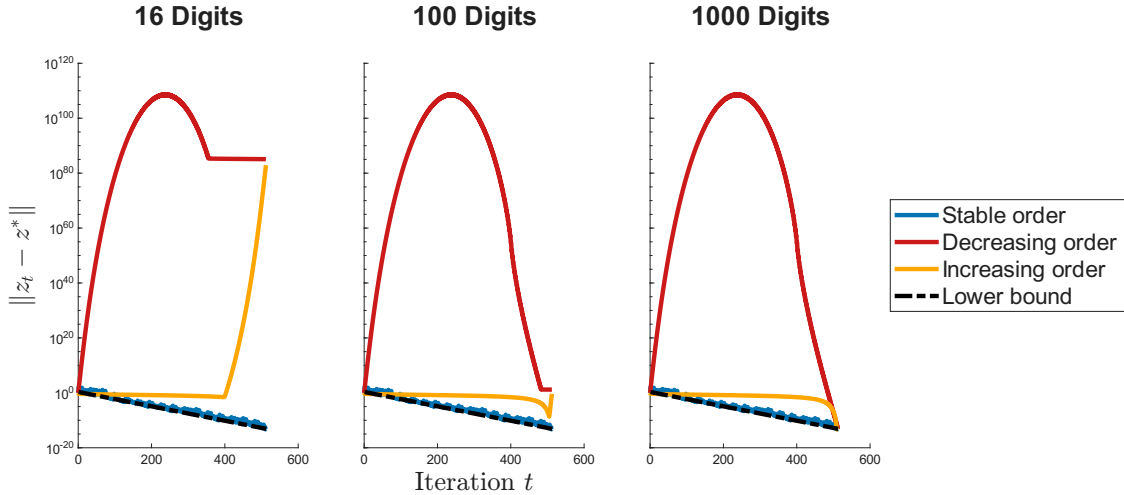


FIGURE 5: Convergence of different orderings of the proposed steps using different levels of numerical precision (“vpa” command in MATLAB). We run 512 iterations on the random bilinear problem described in [Appendix B.5](#).

These types of stability issues are well-studied in the literature on GD stepsize schedules for convex quadratic minimization. In particular, fractal permutations of the Chebyshev stepsize schedule are known to ensure better stability in both theory and practice [\[3, 43, 44, 45, 46, 47\]](#). Because our GDA stepsize schedules emulate these classical GD stepsize schedules (see [Observation 1.1](#) for the precise statement), this immediately suggests the following stable stepsize schedule for GDA.

Definition A.2 (Stable ordering of the slingshot stepsize schedule). *A more numerically stable version of [Definition 3.1](#) is*

$$\alpha_{2t} = -\beta_{2t} = -\alpha_{2t+1} = \beta_{2t+1} = h_{s_t^T}, \quad \forall t = 1, 2, \dots, T. \quad (\text{A.4})$$

Here $\{h_t\}_{t=0}^{T-1}$ are the stepsizes in [Definition 3.1](#), but the difference is in the indexing: s^T is the

stable ordering proposed in [43], namely initialized with $s^1 = [0]$ and defined recursively as

$$s^{2T} := \text{interlace}(s^T, 2T - 1 - s^T), \quad (\text{A.5})$$

where $\text{interlace}([a_1, \dots, a_k], [b_1, \dots, b_k]) := [a_1, b_1, \dots, a_k, b_k]$.

We refer to [3] for a modern exposition of this stable fractal ordering, as well as visualizations, interpretations, and extensions. In practice, this version of the slingshot stepsize schedule appears to be more stable to rounding errors, see Figure 5. And in theory, due to Observation 1.1, one can use known stability results for GD in [3, §3.3] for additive error (which can capture rounding error or inexact gradient computation) in order to prove analogous bounds for GDA.

B Deferred details

B.1 General second-order approximations

Here we generalize the informal derivations in §5 to arbitrary GDA schedules. Let us begin by writing the GDA update more concisely as

$$z_{2t+1} = z_{2t} + \mathbf{D}_{2t} \nabla f(z_{2t}) \quad \text{where} \quad z_{2t} := \begin{bmatrix} x_{2t} \\ y_{2t} \end{bmatrix} \quad \text{and} \quad \mathbf{D}_{2t} := \begin{bmatrix} -\alpha_{2t} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \beta_{2t} \mathbf{I} \end{bmatrix}.$$

By a similar Taylor expansion as in §5, for sufficiently small stepsizes,

$$\nabla f(z_{2t+1}) = \nabla f(z_{2t} + \mathbf{D}_{2t} \nabla f(z_{2t})) \approx z_{2t} + \nabla^2 f(z_{2t}) \mathbf{D}_{2t} \nabla f(z_{2t}).$$

Hence two steps of GDA is approximately equal to

$$\begin{aligned} z_{2t+2} &= z_{2t+1} + \mathbf{D}_{2t+1} \nabla f(z_{2t+1}) \\ &\approx z_{2t} + (\mathbf{D}_{2t} + \mathbf{D}_{2t+1}) \nabla f(z_{2t}) + \mathbf{D}_{2t+1} \nabla^2 f(z_{2t}) \mathbf{D}_{2t} \nabla f(z_{2t}). \end{aligned} \quad (\text{B.1})$$

This recovers the derivations in §5.1 and §5.2 by plugging in the corresponding slingshot stepsize schedules (and taking an expectation for the latter scheme as it is randomized).

To interpret the expansion (B.1), notice that the first movement term $(\mathbf{D}_{2t} + \mathbf{D}_{2t+1}) \nabla f(z_{2t})$ amounts to a single GDA step from z_{2t} , with stepsizes $\alpha_{2t} + \alpha_{2t+1}$ and $\beta_{2t} + \beta_{2t+1}$ in the x and y variables, respectively. Recall from §1.2 that property (iii) of slingshot stepsize schedules requires positive net movement $\alpha_{2t} + \alpha_{2t+1}, \beta_{2t} + \beta_{2t+1} \geq 0$ in both variables; thus this GDA movement in (B.1) amounts to standard “forward” net movement in each variable. The other movement term $\mathbf{D}_{2t+1} \nabla^2 f(z_{2t}) \mathbf{D}_{2t} \nabla f(z_{2t})$ is related to Hamiltonian gradient descent; this connection is precise when $\mathbf{D}_{2t+1}$ and $\mathbf{D}_{2t}$ are scalings of the identity (in expectation), as is the case for the versions of the slingshot stepsize schedule discussed in §5.2.

B.2 Proof of Lemma 2.2

(i) **Constant stepsizes.** In this case, GDA has linear time-invariant dynamics:

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \mathbf{U} \begin{bmatrix} x_t \\ y_t \end{bmatrix} \quad \text{where} \quad \mathbf{U} = \begin{bmatrix} \mathbf{I} & -\alpha_t \mathbf{B} \\ \beta_t \mathbf{B}^\top & \mathbf{I} \end{bmatrix}. \quad (\text{B.2})$$

Let $\mathbf{B} = \mathbf{V}\mathbf{\Sigma}\mathbf{W}^\top$ denote the singular value decomposition of $\mathbf{B}$, and let $\sigma_1, \dots, \sigma_k$ denote the non-zero singular values. Then, for an appropriate permutation matrix $\mathbf{\Pi}$,

$$\mathbf{U} = \begin{bmatrix} \mathbf{V} & \mathbf{0} \\ \mathbf{0} & \mathbf{W} \end{bmatrix} \mathbf{\Pi} \mathbf{D} \mathbf{\Pi}^\top \begin{bmatrix} \mathbf{V}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{W}^\top \end{bmatrix}$$

where $\mathbf{D} = \mathbb{D}(\mathbf{M}_1, \dots, \mathbf{M}_k, \mathbf{I})$ is the diagonal concatenation of the 2×2 matrices

$$\mathbf{M}_j := \begin{bmatrix} 1 & -\alpha\sigma_j \\ \beta\sigma_j & 1 \end{bmatrix}$$

and an identity matrix of the appropriate dimension $d_x + d_y - 2k$. Since the spectral radius is invariant with respect to similarity transforms, and since also the spectral radius of a diagonal concatenation is equal to the largest spectral radius among the constituent diagonal components, it follows that

$$\rho(\mathbf{U}) = \max\{\rho(\mathbf{M}_1), \dots, \rho(\mathbf{M}_k), \rho(\mathbf{I})\} = \max\{|1 \pm i\sigma_1\sqrt{\alpha\beta}|, \dots, |1 \pm i\sigma_k\sqrt{\alpha\beta}|, 1\} \geq 1, \quad (\text{B.3})$$

where the last inequality is strict if α, β are nonzero. Hence GDA with constant stepsizes does not converge on the bilinear min-max problem (3.1), and moreover diverges if α, β are nonzero.

(ii) Non-negative stepsizes. We proceed by a nearly identical diagonalization argument as above, except now the dynamics of GDA are potentially *time-varying*, with update matrices

$$\mathbf{U}_t = \begin{bmatrix} \mathbf{V} & \mathbf{0} \\ \mathbf{0} & \mathbf{W} \end{bmatrix} \mathbf{\Pi} \mathbf{D}_t \mathbf{\Pi}^\top \begin{bmatrix} \mathbf{V}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{W}^\top \end{bmatrix}$$

where $\mathbf{D}_t = \mathbb{D}(\mathbf{M}_{1,t}, \dots, \mathbf{M}_{k,t}, \mathbf{I})$ and

$$\mathbf{M}_{j,t} := \begin{bmatrix} 1 & -\alpha_t\sigma_j \\ \beta_t\sigma_j & 1 \end{bmatrix}.$$

Since the determinant is invariant with respect to orthogonal transformations and permutations, and since also the determinant of a diagonal concatenation equals the product of the determinants of the constituent diagonal components, it follows that

$$\det(\mathbf{U}_t) = \det(\mathbf{D}_t) = \prod_{j=1}^k \det(\mathbf{M}_{j,t}) = \prod_{j=1}^k (1 + \alpha_t\beta_t\sigma_j^2) \geq 1,$$

where the last inequality is strict if α_t, β_t are strictly positive. Since the spectral radius of a matrix is lower bounded by the geometric mean of the eigenvalues, this implies that $\|\prod_{t=1}^T \mathbf{U}_t\| \geq \rho(\prod_{t=1}^T \mathbf{U}_t) \geq \prod_{t=1}^T \det(\mathbf{U}_t)^{1/(m+n)} \geq 1$. Thus GDA with non-negative stepsizes does not converge on the bilinear min-max problem (3.1), and moreover diverges if α_t, β_t are strictly positive.

(iii) Symmetric stepsizes. The argument is identical to case (ii), with one modification: the reason that $\det(\mathbf{M}_{j,t}) \geq 1$ is now due to $\alpha_t = \beta_t \in \mathbb{R}$ rather than $\alpha_t, \beta_t \geq 0$.

B.3 Proof of Theorem 3.4

This lower bound for *asymmetric* Krylov-subspace algorithms follows by a straightforward adaptation of the known lower bounds for *symmetric* Krylov-subspace algorithms [8, 36]. We begin by recalling the definition of these classes of algorithms. Symmetric Krylov-subspace algorithms [8, 36] are iterative algorithms that satisfy the “linear span assumption”

$$\begin{bmatrix} x_t \\ y_t \end{bmatrix} \in \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \text{Span} \left\{ \begin{bmatrix} -\nabla_x f(x_s, y_s) \\ \nabla_y f(x_s, y_s) \end{bmatrix} \right\}_{s \in \{0, \dots, t-1\}}.$$

This captures many common algorithms, but precludes asymmetric updates to the minimization variable x and the maximization variable y , and thus does not capture the slingshot stepsize schedule (among other algorithms). Such updates are captured by the following richer class of algorithms.

Definition B.1 (Asymmetric Krylov-subspace algorithms). *An iterative algorithm for a min-max optimization problem is an asymmetric Krylov-subspace algorithm if for all t , its t -th iterate satisfies*

$$\begin{aligned} x_t &\in x_0 + \text{Span}\{\nabla_x f(x_s, y_s)\}_{s \in \{0, \dots, t-1\}}, \\ y_t &\in y_0 + \text{Span}\{\nabla_y f(x_s, y_s)\}_{s \in \{0, \dots, t-1\}}. \end{aligned}$$

We now turn to proving Theorem 3.4. Briefly, the argument reduces the question of lower bounding the convergence rate of symmetric Krylov-subspace algorithms for bilinear min-max optimization problems, to lower bounding the maximal value of a low-degree polynomial over an interval—a standard result for lower bounds in quadratic optimization (see, e.g., [66]). In the asymmetric case, this reduction is made possible by the following lemma, which can be shown by induction; see [87, Appendix C.1]. Note that this is slightly more involved than the symmetric case, because in that case the 2×2 block matrix of polynomials in this lemma can be simplified to a polynomial of the 2×2 block matrix $\begin{bmatrix} I & B \\ -B^T & I \end{bmatrix}$.

Lemma B.2. *For every iteration t , the t -th iterate of an asymmetric Krylov-subspace algorithm on the bilinear problem (3.1) is of the form*

$$\begin{bmatrix} x_t - x^* \\ y_t - y^* \end{bmatrix} = \begin{bmatrix} p^{(11)}(BB^T) & p^{(12)}(BB^T)B \\ p^{(21)}(B^T B)B^T & p^{(22)}(B^T B) \end{bmatrix} \begin{bmatrix} x_0 - x^* \\ y_0 - y^* \end{bmatrix}, \quad (\text{B.4})$$

where $p^{(11)}$ and $p^{(22)}$ are polynomials of degree at most $\lfloor t/2 \rfloor$, $p^{(12)}$ and $p^{(21)}$ are polynomials of degree at most $\lfloor t/2 \rfloor - 1$, and $p^{(11)}(0) = p^{(22)}(0) = 1$.

For simplicity, we prove Theorem 3.4 below for non-adaptive Krylov-subspace algorithms, i.e., algorithms for which the coefficients of the linear combination in Definition B.1 do not depend on the instance. This can be upgraded to adaptive algorithms by appealing to a stronger lower bound for convex quadratic optimization, namely rather than appealing to Lemma 3.2 (which can only be used to show a lower bound for non-adaptive algorithms for convex quadratic optimization), instead appeal to [4, Theorem 5.1] (which shows that the convergence rate R_T is also optimal for adaptive algorithms).⁸ The proof is otherwise the same, as it relies on the same reduction.

Proof of Theorem 3.4. Consider a 1-dimensional bilinear min-max problem $\min_{x \in \mathbb{R}} \max_{y \in \mathbb{R}} b(x - x^*)(y - y^*)$, where $y^* = y_0$, $x^* \neq x_0$, and $b^2 \in [m, M]$ is a parameter to be chosen. By Lemma B.2, $x_{2T} - x^* = p(b^2) \cdot (x_0 - x^*)$ for some polynomial p of degree at most T satisfying $p(0) = 1$. By Lemma 3.2, there exists $\lambda \in [m, M]$ satisfying $|p(\lambda)| \geq R_T$. Thus, by choosing $b := \sqrt{\lambda}$, we have $|x_{2T} - x^*| \geq R_T |x_0 - x^*|$. The result now follows since $|y_{2T} - y^*| \geq 0 = |y_0 - y^*|$ holds trivially. $\square$

⁸The classical lower bound of [64] could also be used in lieu of [4, Theorem 5.1], but that only matches R_T asymptotically and is weaker for every finite T .

B.4 Proof of Lemma 3.6

The extremal polynomial problem in Lemma 3.6 does not directly fall under the standard statement of Chebyshev’s equioscillation theorem (e.g., [83, Theorem 10.1]), but optimality of p follows from a minor modification of that classical argument. Specifically, suppose for contradiction that p is not optimal; then there exists $\tilde{p} \in \mathcal{P}_{2T}$ for which $\tilde{q}(\lambda) := \lambda\tilde{p}(\lambda)$ satisfies $\max_{\lambda \in [-L, L]} |\tilde{q}(\lambda)| < \max_{\lambda \in [-L, L]} |q(\lambda)|$ where $q(\lambda) := \lambda p(\lambda)$. Note that $q(\lambda)$ is proportional to $\mathcal{T}_{2T+1}(\lambda/L)$ and therefore equioscillates $2T + 2$ times on $[-L, L]$ by the trigonometric definition of Chebyshev polynomials (see §2.3). Hence $q - \tilde{q}$ alternates signs at least $2T + 2$ times on $[-L, L]$. Hence $q - \tilde{q}$ has at least $2T + 1$ roots. But q and $\tilde{q}$ have degree at most $2T$, hence so does $q - \tilde{q}$, contradiction.

Now that we have established optimality of p , the claimed alternative expression for p and the value of the extremal problem follow by the definition of Chebyshev polynomials, see §2.3.

B.5 Experimental setup

Figures 3 and 5 are preliminary numerical simulations that illustrate the convergence and stability properties, respectively, of the slingshot stepsize schedule on bilinear problems. The problem instances $\min_x \max_y x^T \mathbf{B} y$ are generated by drawing $\tilde{\mathbf{B}} \in \mathbb{R}^{30 \times 30}$ with i.i.d. uniform entries in $[0, 1]$, and setting $\mathbf{B} = \mathbf{U}(\Sigma + \mathbf{I})\mathbf{V}^T$ where $\tilde{\mathbf{B}} = \mathbf{U}\Sigma\mathbf{V}^T$ denotes the SVD decomposition of $\tilde{\mathbf{B}}$. This ensures that the singular values of $\mathbf{B}$ are lower bounded by $m = 1$, and in the problem instance shown we use an upper bound $M = 300$ (i.e., $L := \sqrt{M} = \sqrt{300}$). Qualitatively similar results are observed in other problems.

The stepsizes for extragradient, optimistic GDA, and extra anchored gradient are respectively set to the standard prescriptions of $1/(\sqrt{2}L)$ [31], $1/(3L)$ [32], and $1/(8L)$ [87]. For a fair comparison, every full step of the extragradient and extra anchored gradient algorithms is counted as two iterations in Figures 3 and 4 because it uses twice as many gradient queries.

In these experiments, the practical performance of the compared algorithms matches their theoretical bounds. For example, the slow convergence of extragradient and optimistic GDA in Figure 3 illustrates the known rate $\mathcal{O}(\kappa \log 1/\varepsilon)$, and the slingshot stepsize schedule converges at the optimally accelerated rate $\mathcal{O}(\sqrt{\kappa} \log 1/\varepsilon)$ shown in Theorem 3.3. As another example, extragradient, optimistic GDA, and the (non-tailored version of) slingshot stepsizes for convex-concave problems match their theoretical rates of $\mathcal{O}(1/\varepsilon)$ in Figure 4, and the convergence of extra anchored gradient matches the accelerated rate of $\mathcal{O}(1/\sqrt{\varepsilon})$ for smooth convex-concave problems.

References

- [1] Negative stepsizes make gradient-descent-ascent converge – verification of identities computer algebra script. <https://jasonaltschuler.github.io/NegativeStepsizes>.
- [2] Jacob Abernethy, Kevin A. Lai, and Andre Wibisono. Last-iterate convergence rates for min-max optimization: convergence of Hamiltonian gradient descent and consensus optimization. In *International Conference on Algorithmic Learning Theory*, 2021.
- [3] Naman Agarwal, Surbhi Goel, and Cyril Zhang. Acceleration via fractal learning rate schedules. In *International Conference on Machine Learning*, 2021.
- [4] Jason M. Altschuler. Greed, hedging, and acceleration in convex optimization. Master’s thesis, Massachusetts Institute of Technology, 2018.
- [5] Jason M Altschuler and Pablo A Parrilo. Acceleration by stepsize hedging: Silver stepsize schedule for smooth convex optimization. *Mathematical Programming*, pages 1–14, 2024.

- [6] Jason M Altschuler and Pablo A Parrilo. Acceleration by random stepsizes: Hedging, equalization, and the Arcsine stepsize schedule. *arXiv preprint arXiv:2412.05790*, 2024.
- [7] Jason M Altschuler and Pablo A Parrilo. Acceleration by stepsize hedging: Multi-step descent and the silver stepsize schedule. *Journal of the ACM*, 72(2):1–38, 2025.
- [8] Waïss Azizian, Damien Scieur, Ioannis Mitliagkas, Simon Lacoste-Julien, and Gauthier Gidel. Accelerating smooth games by manipulating spectral shapes. In *International Conference on Artificial Intelligence and Statistics*, 2020.
- [9] Waïss Azizian, Ioannis Mitliagkas, Simon Lacoste-Julien, and Gauthier Gidel. A tight and unified analysis of gradient-based methods for a whole spectrum of differentiable games. In *International Conference on Artificial Intelligence and Statistics*, 2020.
- [10] David Balduzzi, Sebastien Racaniere, James Martens, Jakob Foerster, Karl Tuyls, and Thore Graepel. The mechanics of n -player differentiable games. In *International Conference on Machine Learning*, 2018.
- [11] Aharon Ben-Tal, Arkadi Nemirovski, and Laurent El Ghaoui. Robust optimization. 2009.
- [12] Michele Benzi, Gene H Golub, and Jörg Liesen. Numerical solution of saddle point problems. *Acta Numerica*, 14:1–137, 2005.
- [13] Dimitri P. Bertsekas. *Constrained Optimization and Lagrange Multiplier Methods*. Academic Press, 2014.
- [14] Jinho Bok and Jason M Altschuler. Accelerating proximal gradient descent via silver stepsizes. *arXiv preprint arXiv:2412.05497*, 2024.
- [15] Vivek S. Borkar. Stochastic approximation with two time scales. *Systems & Control Letters*, 29(5):291–294, 1997.
- [16] Ekaterina Borodich, Georgiy Kormakov, Dmitry Kovalev, Aleksandr Beznosikov, and Alexander Gasnikov. Optimal algorithm with complexity separation for strongly convex-strongly concave composite saddle point problems. *arXiv preprint arXiv:2307.12946*, 2023.
- [17] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends® in Machine Learning*, 3(1):1–122, 2011.
- [18] Jiseok Chae, Kyuwon Kim, and Donghwan Kim. Two-timescale extragradient for finding local minimax points. In *International Conference on Learning Representations*, 2024.
- [19] Antonin Chambolle and Thomas Pock. A first-order primal-dual algorithm for convex problems with applications to imaging. *Journal of Mathematical Imaging and Vision*, 40(1):120–145, 2011.
- [20] Andrew R Conn, Katya Scheinberg, and Luis N Vicente. *Introduction to derivative-free optimization*. SIAM, 2009.
- [21] Yuhong Dai. Further insight into the convergence of the Fletcher-Reeves method. *Science in China Series A: Mathematics*, 42(9):905–916, 1999.
- [22] Constantinos Daskalakis, Andrew Ilyas, Vasilis Syrgkanis, and Haoyang Zeng. Training GANs with optimism. In *International Conference on Learning Representations*, 2018.
- [23] Tobin A Driscoll, Kim-Chuan Toh, and Lloyd N Trefethen. From potential theory to matrix iterations in six steps. *SIAM Review*, 40(3):547–578, 1998.
- [24] Yoel Drori and Marc Teboulle. Performance of first-order methods for smooth convex minimization: a novel approach. *Mathematical Programming*, 145(1):451–482, 2014.
- [25] Alexandre d’Aspremont, Damien Scieur, and Adrien Taylor. Acceleration methods. *Foundations and Trends® in Optimization*, 5(1-2):1–245, 2021.

- [26] Daniel Gabay and Bertrand Mercier. A dual algorithm for the solution of nonlinear variational problems via finite element approximation. *Computers & Mathematics with Applications*, 2(1):17–40, 1976.
- [27] Gauthier Gidel, Reyhane Askari Hemmat, Mohammad Pezeshki, Rémi Le Priol, Gabriel Huang, Simon Lacoste-Julien, and Ioannis Mitliagkas. Negative momentum for improved game dynamics. In *International Conference on Artificial Intelligence and Statistics*, 2019.
- [28] R. Glowinski and A. Marroco. Sur l’approximation, par éléments finis d’ordre un, et la résolution, par pénalisation-dualité d’une classe de problèmes de dirichlet non linéaires. *Revue française d’automatique, informatique, recherche opérationnelle. Analyse numérique*, 9:41–76, 1975.
- [29] Noah Golowich, Sarath Pattathil, Constantinos Daskalakis, and Asuman Ozdaglar. Last iterate is slower than averaged iterate in smooth convex-concave saddle point problems. In *Conference on Learning Theory*, page 1758–1784, 2020.
- [30] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, 2014.
- [31] Eduard Gorbunov, Nicolas Loizou, and Gauthier Gidel. Extragradient method: $\mathcal{O}(1/k)$ last-iterate convergence for monotone variational inequalities and connections with cocoercivity. In *International Conference on Artificial Intelligence and Statistics*, 2022.
- [32] Eduard Gorbunov, Adrien Taylor, and Gauthier Gidel. Last-iterate convergence of optimistic gradient method for monotone variational inequalities. *Advances in Neural Information Processing Systems*, 35, 2022.
- [33] Baptiste Goujaud, Damien Scieur, Aymeric Dieuleveut, Adrien B Taylor, and Fabian Pedregosa. Super-acceleration with cyclical step-sizes. In *International Conference on Artificial Intelligence and Statistics*, pages 3028–3065, 2022.
- [34] Martin Hast, Karl Johan Åström, Bo Bernhardsson, and Stephen Boyd. PID design by convex-concave optimization. In *European Control Conference*, pages 4460–4465. IEEE, 2013.
- [35] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Advances in Neural Information Processing Systems*, 30, 2017.
- [36] Adam Ibrahim, Waiss Azizian, Gauthier Gidel, and Ioannis Mitliagkas. Linear lower bounds and conditioning of differentiable games. In *International Conference on Machine Learning*, 2020.
- [37] Ruichen Jiang and Aryan Mokhtari. Generalized optimistic methods for convex-concave saddle point problems. *arXiv preprint arXiv:2202.09674*, 2024.
- [38] Yujia Jin, Aaron Sidford, and Kevin Tian. Sharper rates for separable minimax and finite sum optimization via primal-dual extragradient methods. In *Conference on Learning Theory*, 2022.
- [39] Zdeněk Kalousek. Steepest descent method with random step lengths. *Foundations of Computational Mathematics*, 17(2):359–422, 2017.
- [40] Jonathan Kelner, Annie Marsden, Vatsal Sharan, Aaron Sidford, Gregory Valiant, and Honglin Yuan. Big-step-little-step: Efficient gradient methods for objectives with multiple scales. In *Conference on Learning Theory*, pages 2431–2540, 2022.
- [41] Galina M Korpelevich. The extragradient method for finding saddle points and other problems. *Matecon*, 12: 747–756, 1976.
- [42] Dmitry Kovalev and Ekaterina Borodich. On linear convergence in smooth convex-concave bilinearly-coupled saddle-point optimization: Lower bounds and optimal algorithms. *arXiv preprint arXiv:2411.14601*, 2024.
- [43] VI Lebedev and SA Finogenov. Ordering of the iterative parameters in the cyclical Chebyshev iterative method. *USSR Computational Mathematics and Mathematical Physics*, 11(2):155–170, 1971.

- [44] VI Lebedev and SA Finogenov. Solution of the parameter ordering problem in Chebyshev iterative methods. *USSR Computational Mathematics and Mathematical Physics*, 13(1):21–41, 1973.
- [45] VI Lebedev and SA Finogenov. Utilization of ordered Chebyshev parameters in iterative methods. *USSR Computational Mathematics and Mathematical Physics*, 16(4):70–83, 1976.
- [46] VI Lebedev and SA Finogenov. On construction of the stable permutations of parameters for the Chebyshev iterative methods. part I. *Russian Journal of Numerical Analysis and Mathematical Modelling*, 17(5):437–456, 2002.
- [47] VI Lebedev and SA Finogenov. On construction of the stable permutations of parameters for the Chebyshev iterative methods. part II. *Russian Journal of Numerical Analysis and Mathematical Modelling*, 19(3):251–264, 2004.
- [48] Jaewook Lee, Hanseul Cho, and Chulhee Yun. Fundamental benefit of alternating updates in minimax optimization. *arXiv preprint arXiv:2402.10475*, 2024.
- [49] Suheol Lee and Donghwan Kim. Fast extra gradient methods for smooth structured nonconvex-nonconcave minimax problems. *Advances in Neural Information Processing Systems*, 34:22588–22600, 2021.
- [50] Haochuan Li, Farzan Farnia, Subhro Das, and Ali Jadbabaie. On convergence of gradient descent ascent: A tight local analysis. In *International Conference on Machine Learning*, 2022.
- [51] Tengyuan Liang and James Stokes. Interaction matters: A note on non-asymptotic local convergence of generative adversarial networks. In *International Conference on Artificial Intelligence and Statistics*, 2019.
- [52] Tianyi Lin, Chi Jin, and Michael Jordan. On gradient descent ascent for nonconvex-concave minimax problems. In *International Conference on Machine Learning*, 2020.
- [53] Tianyi Lin, Chi Jin, and Michael I. Jordan. Near-optimal algorithms for minimax optimization. In *Conference on Learning Theory*, 2020.
- [54] Tianyi Lin, Chi Jin, and Michael I. Jordan. Two-timescale gradient descent ascent algorithms for nonconvex minimax optimization. *Journal of Machine Learning Research*, 26(11):1–45, 2025.
- [55] Songtao Lu, Rahul Singh, Xiangyi Chen, Yongxin Chen, and Mingyi Hong. Alternating gradient descent ascent for nonconvex min-max problems in robust learning and GANs. In *Asilomar Conference on Signals, Systems, and Computers*, 2019.
- [56] Yulong Lu. Two-scale gradient descent ascent dynamics finds mixed Nash equilibria of continuous games: A mean-field perspective. In *International Conference on Machine Learning*, 2023.
- [57] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations*, 2018.
- [58] John C Mason and David C Handscomb. *Chebyshev polynomials*. Chapman and Hall, 2002.
- [59] Lars Mescheder, Sebastian Nowozin, and Andreas Geiger. The numerics of GANs. In *Advances in Neural Information Processing Systems*, 2017.
- [60] Aryan Mokhtari, Asuman Ozdaglar, and Sarath Pattathil. A unified analysis of extra-gradient and optimistic gradient methods for saddle point problems: Proximal point approach. In *International Conference on Artificial Intelligence and Statistics*, 2020.
- [61] Aryan Mokhtari, Asuman E Ozdaglar, and Sarath Pattathil. Convergence rate of $\mathcal{O}(1/k)$ for optimistic gradient and extragradient methods in smooth convex-concave saddle point problems. *SIAM Journal on Optimization*, 30(4):3230–3251, 2020.
- [62] A. Nedić and A. Ozdaglar. Subgradient methods for saddle-point problems. *Journal of Optimization Theory and Applications*, 142(1):205–228, 2009.

- [63] Arkadi Nemirovski. Prox-method with rate of convergence $\mathcal{O}(1/t)$ for variational inequalities with lipschitz continuous monotone operators and smooth convex-concave saddle point problems. *SIAM Journal on Optimization*, 15(1):229–251, 2004.
- [64] Arkadij Semenovič Nemirovskij and David Borisovich Yudin. Problem complexity and method efficiency in optimization. 1983.
- [65] Yurii Nesterov. A method of solving a convex programming problem with convergence rate $\mathcal{O}(1/k^2)$. *Doklady Akademii Nauk SSSR*, 269(3):543–547, 1983.
- [66] Yurii Nesterov. *Introductory lectures on convex optimization: A basic course*, volume 87. Springer Science & Business Media, 1998.
- [67] Yurii Nesterov. Dual extrapolation and its applications to solving variational inequalities and related problems. *Mathematical Programming*, 109(2):319–344, 2007.
- [68] Yurii Nesterov and Vladimir Spokoiny. Random gradient-free minimization of convex functions. *Foundations of Computational Mathematics*, 17(2):527–566, 2017.
- [69] Samet Oymak. Provable super-convergence with a large cyclical learning rate. *IEEE Signal Processing Letters*, 28:1645–1649, 2021.
- [70] Barak A. Pearlmutter. Fast exact multiplication by the Hessian. *Neural Computation*, 6(1):147–160, 1994. ISSN 0899-7667.
- [71] Boris T Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964.
- [72] L. D. Popov. A modification of the Arrow-Hurwicz method for search of saddle points. *Mathematical notes of the Academy of Sciences of the USSR*, 28(5):845–848, 1980. ISSN 1573-8876.
- [73] Luc Pronzato and Anatoly Zhigljavsky. Gradient algorithms for quadratic optimization with fast convergence rates. *Computational Optimization and Applications*, 50(3):597–617, 2011.
- [74] Alexander Rakhlin and Karthik Sridharan. Online learning with predictable sequences. In *Conference on Learning Theory*, 2013.
- [75] Alexander Rakhlin and Karthik Sridharan. Optimization, learning, and games with predictable sequences. In *Neural Information Processing Systems*, volume 2, 2013.
- [76] Theodore J Rivlin. *Chebyshev polynomials*. Courier Dover Publications, 2020.
- [77] Anne Rubbens, Nizar Bousselmi, Sebastien Colla, and Julien M. Hendrickx. Interpolation constraints for computing worst-case bounds in performance estimation problems. In *Conference on Decision and Control*, page 3015–3022, 2023.
- [78] Ernest K Ryu, Adrien B Taylor, Carolina Bergeling, and Pontus Giselsson. Operator splitting performance estimation: Tight contraction factors and optimal parameter selection. *SIAM Journal on Optimization*, 30(3):2251–2271, 2020.
- [79] Betty Shea and Mark Schmidt. **Don’t be so positive**: Negative step sizes in second-order methods. *arXiv preprint arXiv:2411.11224*, 2024.
- [80] Mikhail V Solodov and Benar F Svaiter. A hybrid approximate extragradient–proximal point algorithm using the enlargement of a maximal monotone operator. *Set-Valued Analysis*, 7(4):323–345, 1999.
- [81] Adrien B Taylor, Julien M Hendrickx, and François Glineur. Smooth strongly convex interpolation and exact worst-case performance of first-order methods. *Mathematical Programming*, 161:307–345, 2017.
- [82] Quoc Tran-Dinh and Yang Luo. Halpern-type accelerated and splitting algorithms for monotone inclusions. *arXiv preprint arXiv:2110.08150*, 2021.
- [83] Lloyd N Trefethen. *Approximation theory and approximation practice*,. SIAM, 2013.

- [84] Paul Tseng. On linear convergence of iterative methods for the variational inequality problem. *Journal of Computational and Applied Mathematics*, 60(1):237–252, 1995.
- [85] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 1947.
- [86] Jun-Kun Wang, Jacob Abernethy, and Kfir Y. Levy. No-regret dynamics in the Fenchel game: a unified framework for algorithmic convex optimization. *Mathematical Programming*, 205(1):203–268, 2024.
- [87] TaeHo Yoon and Ernest K Ryu. Accelerated algorithms for smooth convex-concave minimax problems with $\mathcal{O}(1/k^2)$ rate on squared gradient norm. In *International Conference on Machine Learning*, 2021.
- [88] Taeho Yoon, Jaeyeon Kim, Jaewook J. Suh, and Ernest K. Ryu. Optimal acceleration for minimax and fixed-point problems is not unique. In *International Conference on Machine Learning*, 2024.
- [89] David Young. On Richardson’s method for solving linear systems with positive definite matrices. *Journal of Mathematics and Physics*, 32(1-4):243–255, 1953.
- [90] Guodong Zhang and Yuanhao Wang. On the suboptimality of negative momentum for minimax optimization. In *International Conference on Artificial Intelligence and Statistics*, pages 2098–2106, 2021.
- [91] Guodong Zhang, Xuchan Bao, Laurent Lessard, and Roger Grosse. A unified analysis of first-order methods for smooth games via integral quadratic constraints. *Journal of Machine Learning Research*, 22(103):1–39, 2021.
- [92] Guodong Zhang, Yuanhao Wang, Laurent Lessard, and Roger B. Grosse. Near-optimal local convergence of alternating gradient descent-ascent for minimax optimization. In *International Conference on Artificial Intelligence and Statistics*, 2022.
- [93] Junyu Zhang, Mingyi Hong, and Shuzhong Zhang. On lower iteration complexity bounds for the convex concave saddle point problems. *Mathematical Programming*, 194(1–2):901–935, 2022.
- [94] Anatoly Zhigljavsky, Luc Pronzato, and Elena Bukina. An asymptotically optimal gradient algorithm for quadratic optimization with low computational cost. *Optimization Letters*, 7(6):1047–1059, 2013.
- [95] Martin Zinkevich. Online convex programming and generalized infinitesimal gradient ascent. In *International Conference on Machine Learning*, pages 928–936, 2003.